

[HTML](#)[SEMANTIC](#)[ACCESSIBILITY](#)[DEBUGGING](#)

HTML

BY EXAMPLE

HTML'NI 150 TA AMALIY TOPSHIRIQ ORQALI
0 DAN O'RGANISH

FINAL NASHR • 2026

D1 → D6

Bosqichma-bosqich qiyinlik

16 CHAPTER

Nazariya + Quick Check + mini loyiha

150 CHALLENGE

Amaliy mashq va real integratsiya

5 CAPSTONE

Ko'p sahifali yakuniy loyihalar

```
index.html

<!doctype html>
<html lang="uz">
  <head>
    <meta charset="UTF-8">
    <title>HTML by Example</title>
  </head>
  <body>
    <main>
      <h1>Salom, HTML!</h1>
      <p>Amaliyot bilan o'rgan.</p>
    </main>
  </body>
</html>
```

NAZARIYA • EXAMPLE CODE • QUICK CHECK • MINI LOYIHA
CHALLENGE • JAVOBLAR • VALIDATOR • DEVTOOLS • A11Y

O'zbek tilidagi amaliy HTML darsligi

Final nashr haqida

NIMA O'ZGARDI?

- Professional muqova, chapter checkpoint va optional Stretch mashqlar bilan vizual/pedagogik oqim kuchaytirildi.
- D1-D6 qiyinlik progressiyasi, Quick Check, mini loyiha, debugging va capstone tizimi saqlandi va yanada aniqlandi.
- 30 kunlik o'qish rejasi va 100 ballik capstone baholash rubrikasi qo'shildi.
- WHATWG, MDN va W3C kabi rasmiy manbalar uchun alohida "Keyingi o'qish" bo'limi qo'shildi.
- 001-150 challenge javoblari kitobning eng oxirida qoladi; starter assetlar bilan mashqlarni darhol sinash mumkin.

Kitobdan qanday foydalanish kerak?

- 1. Tushuntirishni o'qing va Example Code ni qo'lda tering.
- 2. Vizual diagramma orqali elementlar o'rtasidagi bog'lanishni ko'ring.
- 3. Quick Check savollariga kitobga qaramasdan javob bering.
- 4. Mini loyihani bajaring; u chapterdagi bilimlarni birlashtiradi.
- 5. Challenge larni ketma-ket bajaring. Javoblar kitobning eng oxirida.
- 6. D5-D6 darajasida validator, DevTools va accessibility tekshiruvini odat qiling.

Ish muhiti

VS Code, Notepad++, Sublime Text yoki oddiy matn muharriri yetarli. Fayllarni .html kengaytmasida saqlang. Chrome, Edge yoki Firefoxda sinang. Murakkab sahifalarda DevTools → Elements orqali brauzer yaratgan DOMni tekshiring. Standart xatolar uchun HTML validator ishlatish tavsiya etiladi.

Birinchi ishga tushadigan fayl

```
<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Birinchi sahifa</title>
</head>
<body>
  <h1>Salom, HTML!</h1>
</body>
</html>
```

Mundarija

Part I - 001-145 bosqichma-bosqich challenge. Part II - 146-150 capstone loyihalar. Appendikslar javoblardan oldin, Part III - barcha javoblar esa kitobning eng oxirida.

CH	CHALLENGE	DARAJA	MAVZU
01	001-010	D1	HTML asoslari
02	011-020	D1	HTML hujjati, head/body va debugging
03	021-030	D2	Sarlavhalar, paragraf va matn bloklari
04	031-040	D2	Semantik matn, kod va maxsus belgilar
05	041-050	D2	Havolalar, fayl yo'llari va navigatsiya
06	051-060	D3	Rasmlar, figure, picture va responsive manbalar
07	061-070	D3	Ro'yxatlar va menyu tuzilmalari
08	071-080	D3	Jadvallar: struktura, birlashtirish va accessibility
09	081-090	D4	Formalar: input, label va foydalanuvchi ma'lumoti
10	091-100	D4	Murakkab formalar, fieldset va umumiy atributlar
11	101-110	D4	Semantic HTML va sahifa arxitekturasi
12	111-118	D5	Audio, video, source va subtitrlar
13	119-126	D5	iframe, object, embed va xavfsiz tashqi kontent
14	127-134	D5	head, metadata, SEO va social preview
15	135-140	D6	Accessibility: semantic, alt, label va klaviatura
16	141-145	D6+	Integration, refaktor va debugging workshop

PART II — YAKUNIY LOYIHALAR

- 146 Shaxsiy portfolio
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- 150 FINAL - Nova Studio

APPENDIKSLAR

Glossary • HTML teglar indeksi • Debugging checklist • 30 kunlik reja • Capstone rubrikasi • Rasmiy manbalar

PART III — JAVOBLAR

Quick Check javoblari + 001-145 yechimlar + 146-150 to'liq ko'p-faylli yechimlar. Bu bo'lim kitobning oxirida.

Qiyinlik legendasi

D1 Starter Bitta yangi tushuncha	D2 Easy 2-3 elementni birlashtirish	D3 Medium Struktura va accessibility
D4 Intermediate Bir nechta forma/semantic ko'nikma	D5 Advanced Media, xavfsizlik, SEO	D6 Hard Refaktor, debugging va integratsiya

PART I

HTMLNI O'RGANISH

001-145 • nazariya → misol → tekshiruv → mini loyiha → challenge

HTML asoslari

DARAJA VA YO'NALISH

- D1 - STARTER • 45-60 min
- Fokus: teg, element, atribut, brauzer

Bu chapterda siz

- HTMLning vazifasini va CSS/JavaScriptdan farqini tushunish
- Element, teg, atribut va nesting tushunchalarini ishlatish
- Oddiy HTML kontentini brauzerda ishga tushirish

Tushuntirish

HTML nima qiladi?

HTML (HyperText Markup Language) web-sahifaning mazmuni va tuzilishini belgilaydi. Brauzer HTML hujjatini o'qib, undagi elementlarni sarlavha, paragraf, rasm, havola, jadval yoki forma kabi ma'noli qismlarga ajratadi. HTML dasturlash tili emas: u shart, sikl yoki hisoblash mexanizmini bermaydi. U kontentga ma'no va struktura beradi.

Teg, element va atribut

`<p>` ochuvchi teg, `</p>` yopuvchi teg, ularning orasidagi matn esa element kontentidir. Ba'zi elementlar `<img>`, `<br>`, `<hr>` kabi void element bo'lib, yopuvchi tegga ega emas. Atribut element haqida qo'shimcha ma'lumot beradi: masalan `<a href="about.html">` ichidagi href havola manzilini belgilaydi.

Nesting va tartib

HTML elementlari bir-birining ichiga joylashtirilishi mumkin. Bu nesting deyiladi. Ichkarida ochilgan element odatda tashqaridagi elementdan oldin yopiladi. Noto'g'ri nesting brauzer tomonidan qisman tuzatilishi mumkin, lekin DOM kutilmagan ko'rinishga keladi. Shuning uchun kodni ichkarilatib yozish o'qishni osonlashtiradi.

Brauzerda tekshirish

HTML faylni .html kengaytmasida saqlang va Chrome, Edge yoki Firefoxda oching. O'zgarishdan keyin faylni saqlab, brauzerni yangilang. Birinchi bosqichda maqsad dizayn emas; elementlar to'g'ri strukturada chiqayotganini tekshirishdir.

Example Code

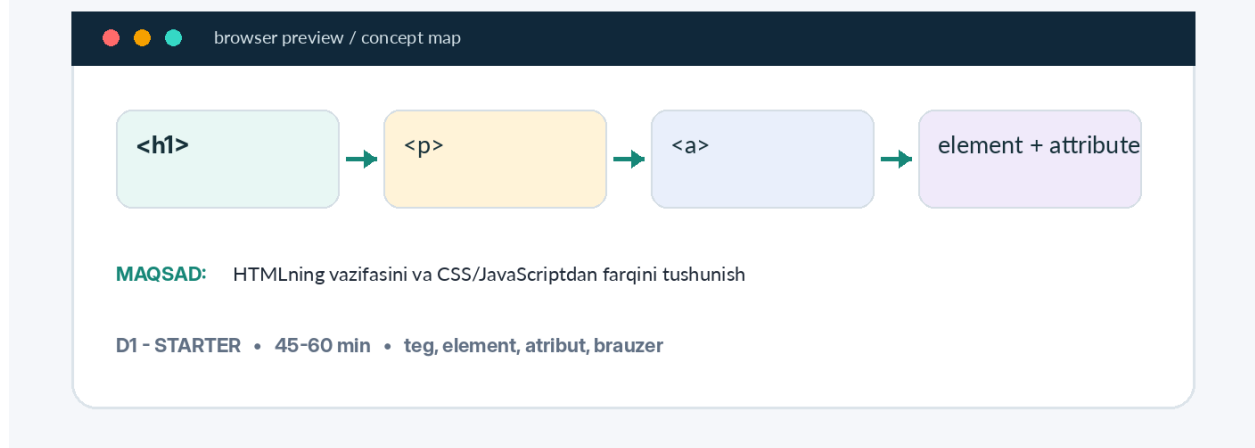
Element anatomiyasi

```
<p class="lead">Salom, HTML!</p>
```

Bir nechta element

```
<h1>Web darsligi</h1>
<p>Bugun HTMLni boshlaymiz.</p>
<hr>
<p>Keyingi mavzu: hujjat tuzilishi.</p>
```

CHAPTER 01 • HTML asoslari



Vizual 1. HTML asoslari - konsept xaritasi

KO'P UCHRAYDIGAN XATOLAR

- Teg nomini xato yozish: `<hl>` o'rniga `<h1>`.
- Yopuvchi teglarni unutish yoki nesting tartibini buzish.
- HTMLni dizayn tili deb o'ylash; dizaynning asosiy qismi CSSga tegishli.

Quick Check

1. HTMLning asosiy vazifasi nima?
2. Element bilan teg o'rtasidagi farq nima?
3. Void elementga ikkita misol keltiring.

MINI LOYIHA

- "Men haqimda" nomli kichik sahifa kontenti yarating: asosiy sarlavha, ikki paragraf, gorizontaal chiziq va HTML izoh ishlating.

Challenges

Avval mustaqil bajaring. Kod ishlagach, semantika va accessibilityni ham tekshiring. Mumkin bo'lgan yechimlar kitobning eng oxiridagi PART III da.

001

D1

Sahifada 'Mening birinchi saytim' degan katta sarlavha chiqaring.

002

D1

Ismingizni paragraf ko'rinishida chiqaring.

003

D1

Bitta h1 va uning ostida ikkita paragraf yarating.

004

D1

Bir paragraf ichida ikki satrni `<br>` yordamida ajrating.

005

D1

Ikki matn bo'limi orasiga gorizontal chiziq qo'ying.

006

D1

HTML izoh yozing: 'Bu sarlavha keyin o'zgartiriladi'.

007

D1

h1, h2 va p ishlatib kichik dars sahifasi yarating.

008

D1

'5 < 10 va 10 > 5' matnini HTML entity yordamida xavfsiz chiqaring.

009

D1

Mualliflik belgisi bilan '2026 HTML Book' yozuvini chiqaring.

010

D1

Faqat o'rgangan teglaringizdan foydalanib mini 'Men haqimda' blokini yarating.

CHAPTER CHECKPOINT • 01 YAKUNLANDI

MEN ENDI:

- ✓ Teg, element va atributni farqlay olaman.
- ✓ Nesting va void elementlarni to'g'ri ishlata olaman.
- ✓ Oddiy HTML kontentini brauzerda tekshira olaman.

STRETCH • JAVOBSIZ

Kitobga qaramasdan "Men haqimda" blokini qayta yozing va bitta ataylab xato nestingni topib tuzating.

HTML hujjati, head/body va debugging

DARAJA VA YO'NALISH

- D1 - STARTER+ • 60-75 min
- Fokus: doctype, head, body, DevTools, validator

Bu chapterda siz

- To'liq HTML5 hujjat skeletini xatosiz yozish
- head va body vazifalarini farqlash
- Oddiy xatolarni brauzer DevTools va validator yordamida topish

Tushuntirish

HTML5 hujjat skeleti

`<!DOCTYPE html>` brauzerga hujjat zamonaviy HTML standartida talqin qilinishini bildiradi. `<html>` ildiz element, `<head>` hujjat metama'lumotlari, `<body>` esa ekranda ko'rinadigan asosiy mazmun uchun ishlatiladi. Bu skeletni boshidan to'g'ri yozish keyingi barcha mavzular uchun tayanch bo'ladi.

Til va kodlash

`<html lang="uz">` sahifaning asosiy tili o'zbekcha ekanini bildiradi. Bu ekran o'quvchi, tarjima vositasi va qidiruv tizimlariga yordam beradi. `<meta charset="UTF-8">` esa o'zbekcha apostrof, kirill, lotin va boshqa belgilarni to'g'ri ko'rsatishga xizmat qiladi.

Title va viewport

`<title>` brauzer tabidagi nom va qidiruv natijasidagi muhim signal hisoblanadi. `<meta name="viewport" content="width=device-width, initial-scale=1.0">` mobil ekranda sahifaning to'g'ri masshtablanishiga yordam beradi. Har real sahifada mazmuniga mos title bo'lishi kerak.

Debugging odati

Kod ishlamasa, tasodifiy o'zgartirish o'rniga tizimli tekshiring: 1) fayl saqlanganmi, 2) teglar yopilganmi, 3) atribut qo'shtirnoqlari juftmi, 4) fayl yo'li to'g'rimi, 5) DevTools Elements panelida DOM qanday qurilgan. HTML Validator ham noto'g'ri nesting va standartga zid markupni topishda foydali.

Example Code

Minimal hujjat

```
<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bosh sahifa</title>
</head>
<body>
  <h1>Salom!</h1>
</body>
</html>
```

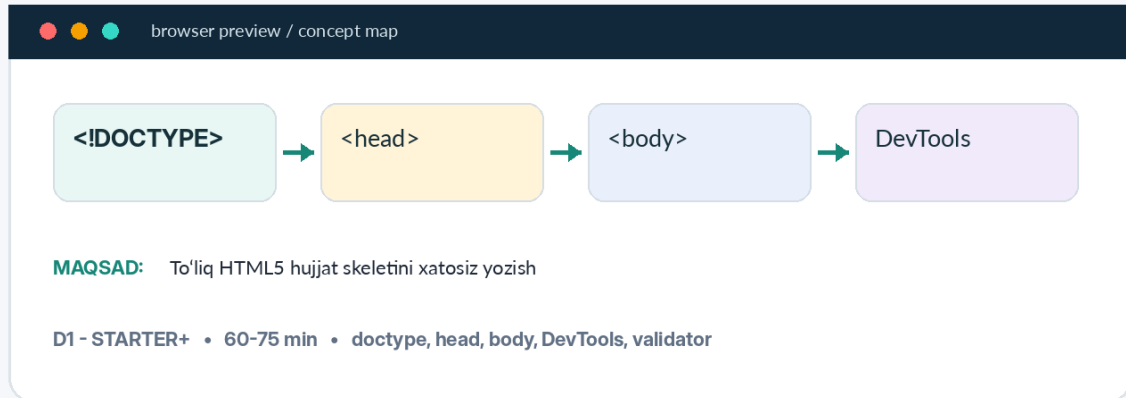
Xatoni topish

```
<!-- Noto'g'ri: title body ichida -->
<body>
  <title>Profil</title>
```

```
</body>

<!-- To'g'ri: title head ichida -->
```

CHAPTER 02 • HTML hujjati, head/body va debugging



Vizual 2. HTML hujjati, head/body va debugging - konsept xaritasi

KO'P UCHRAYDIGAN XATOLAR

- <title> yoki <meta> teglarini body ichiga yozish.
- DOCTYPE yoki charsetni unutish.
- Fayl .txt bo'lib qolganini sezmaslik: masalan index.html.txt.

Quick Check

1. head ichidagi ma'lumotlar odatda kimlar uchun xizmat qiladi?
2. lang="uz" nimaga kerak?
3. HTML xatosini izlashda birinchi uch tekshiruvni sanang.

MINI LOYIHA

- Ataylab 5 ta xato kiritilgan HTML fayl yarating, keyin ularni DevTools/validator yordamida topib to'g'rilang. Xatolar ro'yxatini izoh sifatida yozing.

Challenges

Avval mustaqil bajaring. Kod ishlagach, semantika va accessibilityni ham tekshiring. Mumkin bo'lgan yechimlar kitobning eng oxiridagi PART III da.

011

D1

Eng minimal HTML5 hujjat skeletini yozing.

012

D1

html elementiga o'zbek tili atributini qo'shing.

013

D1

UTF-8 kodlashni head ichida belgilang.

014

D1

Brauzer yorlig'ida "HTML darsligi" nomi chiqadigan hujjat yozing.

015

D1

Mobil qurilmalar uchun viewport meta tegini qo'shing.

016

D1

head ichida title va charset, body ichida h1 bo'lgan hujjat tuzing.

017

D1

body ichidagi kontentni to'g'ri ichkarilatib (indentation) yozing.

018

D1

head ichiga "Muallif: Talaba" degan izoh yozing.

019

D1

doctype yozilmagan hujjatni to'g'rilang.

020

D1

"Mening profilim" nomli to'liq HTML5 sahifa yarating.

CHAPTER CHECKPOINT • 02 YAKUNLANDI**MEN ENDI:**

- ✓ DOCTYPE, html, head va bodyni xatosiz tuza olaman.
- ✓ lang, charset, viewport va titleni to'g'ri joylashtiraman.
- ✓ DevTools/validator bilan oddiy markup xatosini topa olaman.

STRETCH • JAVOBSIZ

5 ta xatoli HTML skelet yozing, 10 daqiqadan keyin unga qaytib barcha xatolarni izoh bilan tuzating.

Sarlavhalar, paragraf va matn bloklari

DARAJA VA YO'NALISH

- D2 - EASY • 60-75 min
- Fokus: h1-h6, p, blockquote, q, pre, address

Bu chapterda siz

- Mantiqiy heading ierarxiasini yaratish
- Paragraf va iqtiboslarni semantik belgilash
- Oldindan formatlangan va aloqa matnlarini to'g'ri yozish

Tushuntirish

Heading ierarxiyasi

h1-h6 elementlari matnni shunchaki kattalashtirmaydi; ular hujjatning mazmuniy ierarxiasini ko'rsatadi. Odatda sahifaning bosh mavzusi h1, uning ichki bo'limlari h2, h2 ichidagi kichik bo'limlar h3 bo'ladi. Darajani tashlab ketish har doim qat'iy xato bo'lmasa-da, mantiqiy daraxtni saqlash o'qish va accessibility uchun yaxshiroq.

Paragraf va satr

<p> alohida fikr yoki paragraf uchun ishlatiladi. Vizual bo'shliq olish uchun ketma-ket
 ishlatish o'rniga mazmunni to'g'ri paragraf va bo'limlarga ajrating.
 manzil, she'r yoki satr uzilishi mazmunning bir qismi bo'lgan joyda ma'noli.

Iqtiboslar

Qisqa iqtibos matn oqimi ichida <q>, alohida katta iqtibos esa <blockquote> bilan belgilanadi. Manba URLini cite atributida ko'rsatish mumkin, lekin foydalanuvchiga ko'rinadigan manba linki alohida berilgani ma'qul.

pre va address

<pre> ichidagi bo'sh joy va satrlar saqlanadi; u ASCII-art, log yoki formatlangan matn uchun qulay. Kodning o'zi uchun ko'pincha <pre><code> juftligi ishlatiladi. <address> sahifa yoki maqola muallifining aloqa ma'lumotlarini belgilash uchun mo'ljallangan.

Example Code

Mantiqiy tuzilma

```
<h1>Web texnologiyalar</h1>
<h2>Frontend</h2>
<h3>HTML</h3>
<p>HTML strukturani belgilaydi.</p>
```

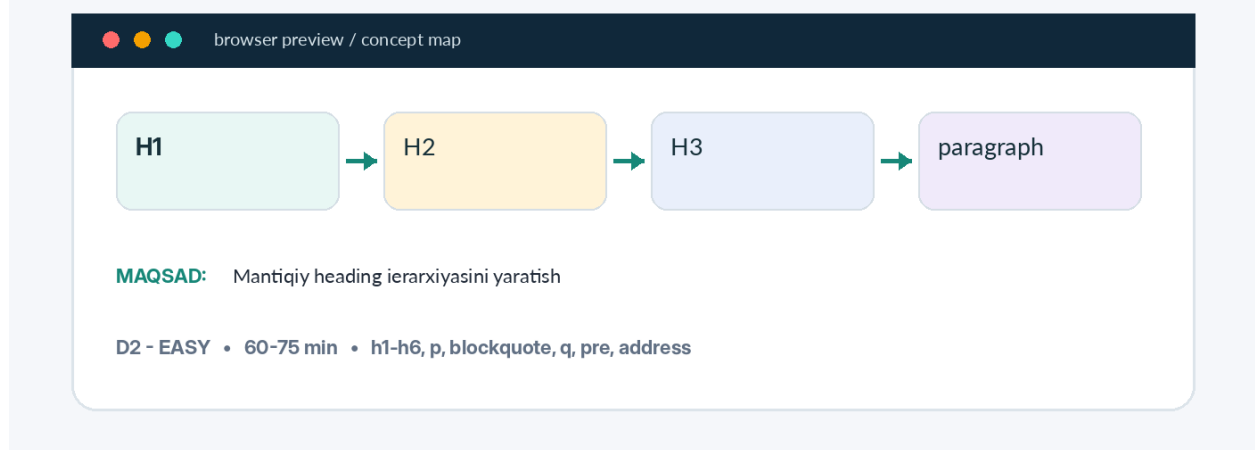
Iqtibos

```
<blockquote cite="https://example.com/source">
  Amaliyot bilimni mustahkamlaydi.
</blockquote>
```

Pre + code

```
<pre><code>&lt;h1&gt;Salom&lt;/h1&gt;</code></pre>
```

CHAPTER 03 • Sarlavhalar, paragraf va matn bloklari



Vizual 3. Sarlavhalar, paragraf va matn bloklari - konsept xaritasi

KO'P UCHRAYDIGAN XATOLAR

- Heading darajasini faqat shrift o'lchami uchun tanlash.
- Har yangi satr uchun `<br>` ishlatish.
- Oddiy kontentning katta qismini `<pre>` ichiga joylash.

Quick Check

1. h1 va h2 o'rtasidagi semantik munosabat qanday?
2. Qachon `<q>`, qachon `<blockquote>` ishlatiladi?
3. `<pre>` bo'sh joylarga qanday munosabatda bo'ladi?

MINI LOYIHA

- 3 bo'limli mini maqola tuzing: h1, kamida ikki h2, bitta h3, iqtibos, ikki paragraf va muallif address elementi bo'lsin.

Challenges

Avval mustaqil bajaring. Kod ishlagach, semantika va accessibilityni ham tekshiring. Mumkin bo'lgan yechimlar kitobning eng oxiridagi PART III da.

021

D2

h1 dan h3 gacha ketma-ket uch darajali sarlavha yarating.

022

D2

Ikki alohida paragrafda HTML va CSS haqida bittadan gap yozing.

023

D2

h1-h6 teglarining barchasini bir marta ishlatib.

024

D2

Qisqa iqtibosni `<q>` bilan yozing.

025

D2

Uzun iqtibos uchun blockquote ishlatish.

026

D2

pre yordamida uch satrli kodga o'xshash matn chiqarish.

027

D2

address tegi yordamida aloqa ma'lumotini yozish.

028

D2

Maqola sarlavhasi, kichik bo'lim va ikkita paragrafdan iborat tuzilma yaratish.

029

D2

Ikki bo'limni <hr> bilan ajratish va har biriga h2 qo'yish.

030

D2

Sarlavhalar ierarxiyasiga amal qilgan mini maqola yaratish.

CHAPTER CHECKPOINT • 03 YAKUNLANDI**MEN ENDI:**

- ✓ Heading ierarxiyasini mantiqan qura olaman.
- ✓ p, q, blockquote, pre va addressni vazifasiga mos tanlay olaman.
- ✓ Matn strukturasi bilan vizual ko'rinishni adashtirmayman.

STRETCH • JAVOBSIZ

Mahalliy yangilik uchun h1/h2/h3, iqtibos va muallif contacti bor mini maqola markupini tuzish.

Semantik matn, kod va maxsus belgilar

DARAJA VA YO'NALISH

- D2 - EASY • 60-75 min
- Fokus: strong, em, mark, code, kbd, entity

Bu chapterda siz

- Muhimlik va urg'uni ma'noli elementlar bilan belgilash
- Formulalar va tahrir izlarini HTMLda ifodalash
- Kod va maxsus belgilarni xavfsiz ko'rsatish

Tushuntirish

Ma'no va ko'rinish

`<strong>` jiddiy muhimlikni, `<em>` urg'uni bildiradi. `<b>` va `<i>` ham ishlatilishi mumkin, lekin ularning semantik ma'nosi boshqacha. Elementni "qalin ko'rinsin" degan sabab bilan emas, matndagi ma'noga qarab tanlash yaxshi HTMLning asosidir.

Tahrir va ajratish

`<mark>` kontekstda ajratilgan matnni, `<del>` o'chirilgan mazmunni, `<ins>` qo'shilgan mazmunni ko'rsatadi. Narx o'zgarishi, hujjat tahriri yoki qidiruv natijasidagi topilgan so'zni belgilash kabi holatlarda foydali.

Formula va texnik matn

`<sub>` pastki indeks, `<sup>` yuqori indeks uchun ishlatiladi. `<code>` kod fragmenti, `<kbd>` foydalanuvchi bosadigan tugma yoki buyruq, `<samp>` dastur chiqishi, `<var>` o'zgaruvchi nomi kabi texnik mazmunga ma'no beradi.

HTML entity

`<` va `>` kabi belgilar HTML sintaksisiga kiradi. Ularni matn sifatida ko'rsatish uchun `<` va `>` yoziladi. `&` ampersand, `©` mualliflik belgisini chiqaradi. Bu ayniqsa HTML darsligida kodni matn sifatida ko'rsatishda muhim.

Example Code

Semantik urg'u

```
<p><strong>Diqqat:</strong> faylni <em>saqlang</em>.</p>
```

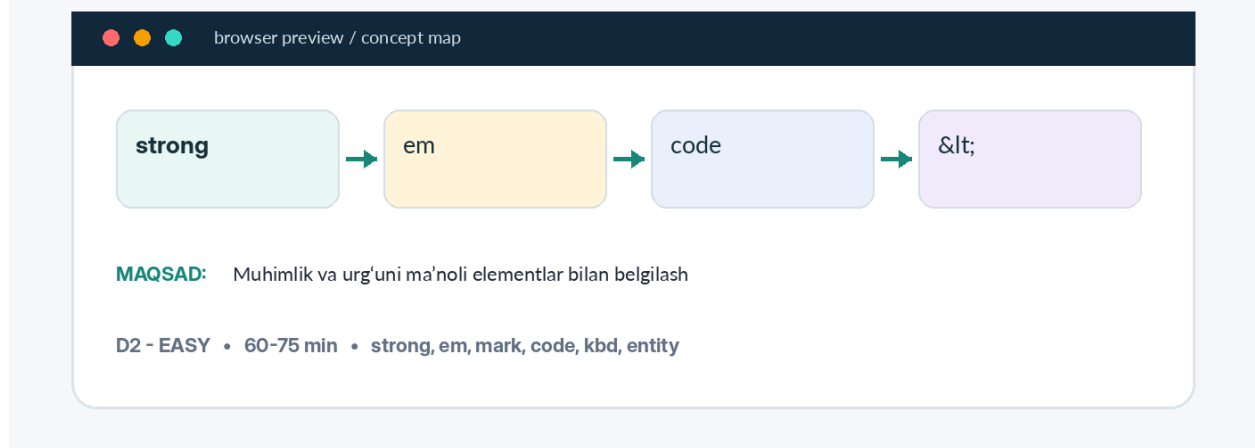
Formula

```
<p>H<sub>2</sub>O va E = mc<sup>2</sup></p>
```

Kod matn sifatida

```
<p><code>&lt;h1&gt;</code> asosiy sarlavha tegidir.</p>
```

CHAPTER 04 • Semantik matn, kod va maxsus belgilar



Vizual 4. Semantik matn, kod va maxsus belgilar - konsept xaritasi

KO'P UCHRAYDIGAN XATOLAR

- strong/em elementlarini faqat bezak uchun ko'p ishlatish.
- <sub>/<sup> ni oddiy kichik shrift o'rnida ishlatish.
- HTML kodini entityga aylantirmasdan <code> ichida yozish.

Quick Check

1. strong bilan b ning asosiy farqi nima?
2. va <ins> qaysi holatda foydali?
3. Nega <code><p></code> yozilganda < ishlatiladi?

MINI LOYIHA

- Mahsulot yangilanishi haqidagi paragraf yozing: eski va yangi narx, muhim ogohlantirish, klaviatura tugmasi va HTML kod namunasi birgalikda bo'lsin.

Challenges

Avval mustaqil bajaring. Kod ishlagach, semantika va accessibilityni ham tekshiring. Mumkin bo'lgan yechimlar kitobning eng oxiridagi PART III da.

031

D2

"Diqqat" so'zini strong bilan muhim qilib belgilang.

032

D2

Bir gapdagi bitta so'zga em orqali urg'u bering.

033

D2

mark yordamida kalit so'zni ajrating.

034

D2

Eski narxni del, yangi narxni ins bilan ko'rsating.

035

D2

H₂O formulasini to'g'ri yozing.**036**

D2

x ning kvadratini HTMLda yozing.

037

D2

Kichik izohni small tegi bilan yozing.

038

D2

Ctrl+S tugmalarini kbd tegi bilan belgilang.

039

D2

HTML tegini matn sifatida code ichida ko'rsating.

040

D2

Bir paragrafda strong, em, mark va code ishlatang.

CHAPTER CHECKPOINT • 04 YAKUNLANDI**MEN ENDI:**

- ✓ strong/em va b/i ma'no farqini tushunaman.
- ✓ code/kbd/samp/var kabi texnik matn elementlarini taniyman.
- ✓ HTML entity yordamida maxsus belgilarni xavfsiz chiqaraman.

STRETCH • JAVOBSIZ

Mahsulot versiya yangilanishi uchun eski/yangi qiymat, warning, klaviatura shortcut va kod fragmenti bor changelog yozing.

Havolalar, fayl yo‘llari va navigatsiya

DARAJA VA YO‘NALISH

- D2 - EASY+ • 75-90 min
- Fokus: a, href, relative path, anchor, nav

Bu chapterda siz

- Ichki va tashqi havolalarni to‘g‘ri qurish
- Relative va absolute manzillarni farqlash
- Sahifa ichidagi va sayt bo‘ylab navigatsiyani tashkil qilish

Tushuntirish

Havolaning asosiy sintaksisi

`<a>` elementining `href` atributi maqsad manzilini belgilaydi. Havola matni foydalanuvchiga qayerga borishini tushuntirishi kerak; “bu yerga bosing” kabi noaniq matn o‘rniga “Kurs dasturini ko‘rish” aniqroq.

Relative va absolute yo‘l

Bir loyiha ichida `about.html`, `pages/contact.html` yoki `../index.html` kabi relative path ishlatiladi. Tashqi resurs uchun `https://` bilan boshlanuvchi to‘liq URL kerak. Papka strukturasi o‘zgarsa relative path ham moslashtiriladi; broken linklarning katta qismi aynan yo‘l xatosidan keladi.

Yangi tab va protokollar

`target="_blank"` havolani yangi tabda ochadi. Tashqi havolada `rel="noopener"` ishlatish yaxshi xavfsizlik amaliyotidir. Email uchun `mailto:`, telefon uchun `tel:` ishlatiladi. Fayl linkida `download` atributi brauzerga yuklab olish niyatini bildiradi, lekin server siyosati yakuniy xatti-harakatga ta‘sir qilishi mumkin.

Anchor va navigatsiya

Sahifa ichida id berilgan elementga `#id` orqali o‘tiladi. `<nav>` asosiy navigatsiya bloklarini semantik belgilaydi. Katta sayt uchun navigatsiyani barcha sahifalarda bir xil tartibda saqlash foydalanuvchiga yo‘l topishni osonlashtiradi.

Example Code

Ichki va tashqi link

```
<a href="about.html">Biz haqimizda</a>
<a href="https://example.com" target="_blank" rel="noopener">Manba</a>
```

Anchor

```
<a href="#contact">Kontaktga o‘tish</a>
...
<h2 id="contact">Kontakt</h2>
```

Nav

```
<nav aria-label="Asosiy navigatsiya">
  <a href="index.html">Bosh sahifa</a>
  <a href="courses.html">Kurslar</a>
</nav>
```

CHAPTER 05 • Havolalar, fayl yo'llari va navigatsiya



Vizual 5. Havolalar, fayl yo'llari va navigatsiya - konsept xaritasi

KO'P UCHRAYDIGAN XATOLAR

- Fayl yo'lini loyiha papkasiga mos yozmaslik.
- Havola matnini maqsadsiz "click here" qilish.
- Yangi tab ochib rel="noopener" ni e'tiborsiz qoldirish.

Quick Check

1. ../ belgisi relative pathda nimani anglatadi?
2. Sahifa ichidagi anchor qaysi atribut bilan bog'lanadi?
3. Nav ichiga har bir linkni joylash shartmi?

MINI LOYIHA

- 3 sahifali mini sayt rejasini yarating: index, about, contact. Har uchallasida bir xil nav bo'lsin, indexda "kontaktga sakrash" anchor ham ishlasin.

Challenges

Avval mustaqil bajaring. Kod ishlagach, semantika va accessibilityni ham tekshiring. Mumkin bo'lgan yechimlar kitobning eng oxiridagi PART III da.

041

D2

about.html sahifasiga havola yarating.

042

D2

example.com saytini yangi tabda ochadigan havola yozing.

043

D2

contact.html ichki sahifasiga "Bog'lanish" linkini yarating.

044

D2

Email yuborish uchun mailto havolasi yozing.

045

D2

Telefon raqamiga tel havolasi yozing.

046

D2

Sahifaning "services" bo'limiga sakraydigan link yarating.

047

D2

files/guide.pdf fayliga havola yarating.

048

D2

PDFni yuklab olishga mo'ljallangan link yozing.

049

D2

Bosh sahifa, Kurslar va Kontakt linklaridan iborat nav bloki yarating.

050

D2

Yuqorida "Pastga" linki, pastda esa "Yuqoriga" linki bo'lgan sahifa bo'lagi yarating.

CHAPTER CHECKPOINT • 05 YAKUNLANDI**MEN ENDI:**

- ✓ Relative va absolute URLni farqlay olaman.
- ✓ Anchor, mailto, tel va download linklarni ishlata olaman.
- ✓ Kichik sayt navigatsiyasida broken pathlarni tekshira olaman.

STRETCH • JAVOBSIZ

pages/contact.html joylashgan ichki papkali 3 sahifali sayt uchun barcha nav yo'llarini o'zingiz rejalang va yozing.

Rasmlar, figure, picture va responsive manbalar

DARAJA VA YO'NALISH

- D3 - MEDIUM • 90 min
- Fokus: img, alt, figure, picture, srcset, lazy

Bu chapterda siz

- Rasmni alt, width va height bilan to'g'ri qo'shish
- figure/figcaption bilan rasmga ma'no berish
- picture/source yordamida mos rasm manbasini tanlash

Tushuntirish

img va alt

`<img>` void elementidir. `src` rasm faylini ko'rsatadi, `alt` esa rasm ko'rinmaganda yoki ekran o'quvchi ishlatilganda uning ma'nosini beradi. Mazmunli rasmda `alt` qisqa va vazifaga mos bo'lsin; dekorativ rasm uchun `alt=""` ekran o'quvchiga uni chetlab o'tishni bildiradi.

O'lcham va layout barqarorligi

`width` va `height` atributlari rasmning tabiiy nisbatiga mos berilsa brauzer rasm yuklanishidan oldin joy ajratadi va layout siljishini kamaytiradi. Rasmni buzib cho'zish kerak emas. Responsive ko'rinish odatda CSS bilan boshqariladi, ammo HTML manba tanlashda muhim rol o'ynaydi.

figure va figcaption

Rasm, diagramma, kod skrinshoti kabi mustaqil kontent `<figure>` ichida guruhlanishi mumkin. `<figcaption>` shu kontentga ko'rinadigan izoh beradi. `Alt` va `caption` bir xil vazifa emas: `alt` rasmning ma'nosini o'rnini bosadi, `caption` esa foydalanuvchiga qo'shimcha kontekst beradi.

picture, source va lazy loading

`<picture>` ichida bir nechta `<source>` berib WebP kabi formatni yoki ekran o'lchamiga mos rasmni tanlash mumkin. Oxirida `<img>` fallback bo'lishi shart. `loading="lazy"` ekranning pastidagi rasmlarni keyinroq yuklashga yordam beradi; yuqoridagi asosiy hero rasmda uni har doim ishlatish kerak emas.

Example Code

Mazmunli rasm

```

```

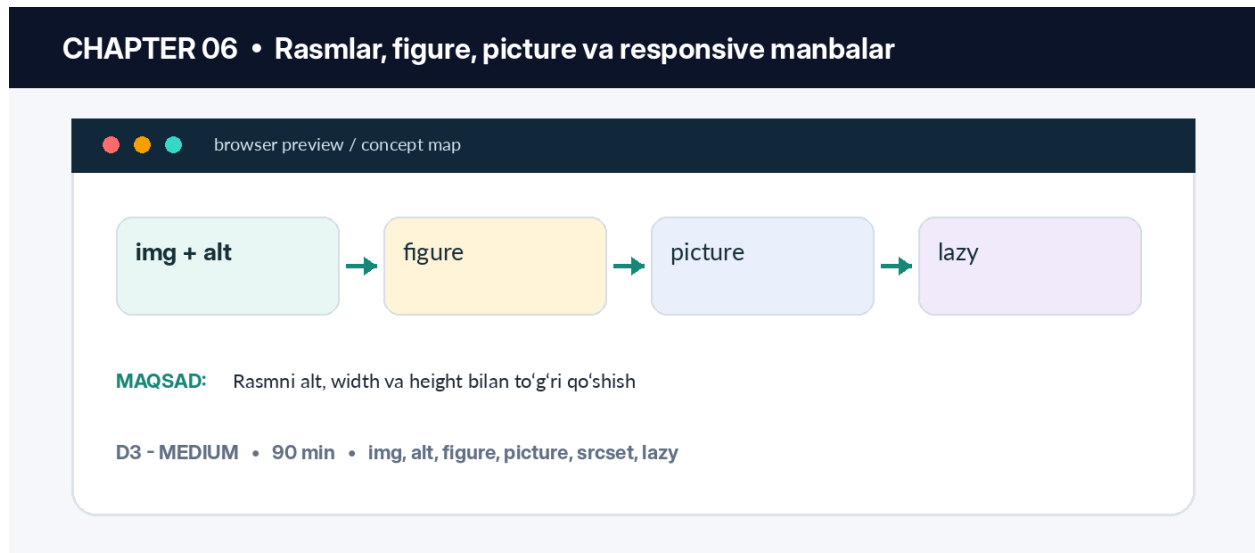
Figure

```
<figure>
  
  <figcaption>1-rasm. Yillik o'sish.</figcaption>
</figure>
```

Picture

```
<picture>
  <source srcset="images/hero.webp" type="image/webp">
  
</picture>
```

CHAPTER 06 • Rasmlar, figure, picture va responsive manbalar



Vizual 6. Rasmlar, figure, picture va responsive manbalar - konsept xaritasi

KO'P UCHRAYDIGAN XATOLAR

- Mazmunli rasmda altni tashlab ketish yoki “rasm” deb yozish.
- width va height bilan nisbatni buzish.
- picture ichida fallback img qoldirmaslik.

Quick Check

1. Dekorativ rasmning alt qiymati qanday bo'ladi?
2. figure va alt bir-birini to'liq almashtiradimi?
3. picture ichidagi img nima uchun kerak?

MINI LOYIHA

- Uch rasmdan iborat galereya yarating: har rasm figure/figcaption ichida, pastdagi rasmlarda lazy loading, bir rasmda WebP fallback picture ishlasin.

Challenges

Avval mustaqil bajaring. Kod ishlagach, semantika va accessibilityni ham tekshiring. Mumkin bo'lgan yechimlar kitobning eng oxiridagi PART III da.

051

D3

images/logo.png rasmini alt matn bilan qo'shing.

052

D3

Rasmga width=600 va height=400 atributlarini qo'shing.

053

D3

Dekorativ rasm uchun bo'sh alt yozing.

054

figure va figcaption yordamida rasmga izoh bering.

D3

055

D3

images papkasidagi avatar.jpg rasmini profil sarlavhasi ostiga qo'ying.

056

D3

Rasmni bosganda home.html ga o'tadigan qiling.

057

D3

Bir xil rasmning WebP manbasini picture orqali ustun qo'ying.

058

D3

Kichik va katta ekran uchun ikki source ishlatilgan picture yozing.

059

D3

Rasmni lazy loading bilan yozing.

060

D3

Uchta figure elementidan iborat mini galereya tuzing.

CHAPTER CHECKPOINT • 06 YAKUNLANDI

MEN ENDI:

- ✓ Mazmunli va dekorativ rasm altini farqlay olaman.
- ✓ figure/figcaption va picture/source vazifasini tushunaman.
- ✓ Lazy loadingni qayerda ishlatish kerakligini bilaman.

STRETCH • JAVOBSIZ

Desktop/mobile manbalari va WebP fallbacki bor 4 rasimli accessible galereya markupini tuzing.

Ro'yxatlar va menyu tuzilmalari

DARAJA VA YO'NALISH

- D3 - MEDIUM • 75-90 min
- Fokus: ul, ol, li, dl, nested list

Bu chapterda siz

- Tartibli va tartibsiz ro'yxatni maqsadga qarab tanlash
- Ichma-ich ro'yxatlarni to'g'ri nesting qilish
- Ta'rif ro'yxati va nav menyusini semantik tuzish

Tushuntirish

ul va ol

`<ul>` tartibi muhim bo'lmagan elementlar uchun, `<ol>` esa qadamlar, reyting yoki ketma-ketlik uchun ishlatiladi. Har bir element `<li>` ichida bo'ladi. Ro'yxatni shunchaki nuqta ko'rinishi kerakligi uchun emas, mazmun haqiqatan ro'yxat bo'lganda ishlatiladi.

ol atributlari

`<ol start="5">` sanashni 5 dan boshlaydi, `reversed` esa teskari tartibda raqamlaydi. `value` atributi ayrim li elementlari raqamini o'zgartirishi mumkin. Bular mazmunning raqamli ketma-ketligi haqiqatan muhim bo'lganda foydali.

Ichma-ich ro'yxat

Ichki `<ul>` yoki `<ol>` tegini tegishli `<li>` ichiga joylashtiring. Bu kurs dasturi, kategoriya daraxti yoki ko'p darajali menyu kabi tuzilmalar uchun mos. Juda chuqur nesting foydalanuvchini chalg'itishi mumkin.

dl, dt va dd

Termin va uning ta'rif, savol-javob juftligi yoki metadata juftliklari uchun `<dl>` ishlatiladi. `<dt>` termin yoki kalit, `<dd>` esa uning ta'rif/qiymati. Navigatsiya linklari esa ko'pincha nav ichidagi ul/li tuzilmasi bilan beriladi.

Example Code

Oddiy ro'yxatlar

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
</ul>
<ol>
  <li>Fayl yarating</li>
  <li>Kodni yozing</li>
</ol>
```

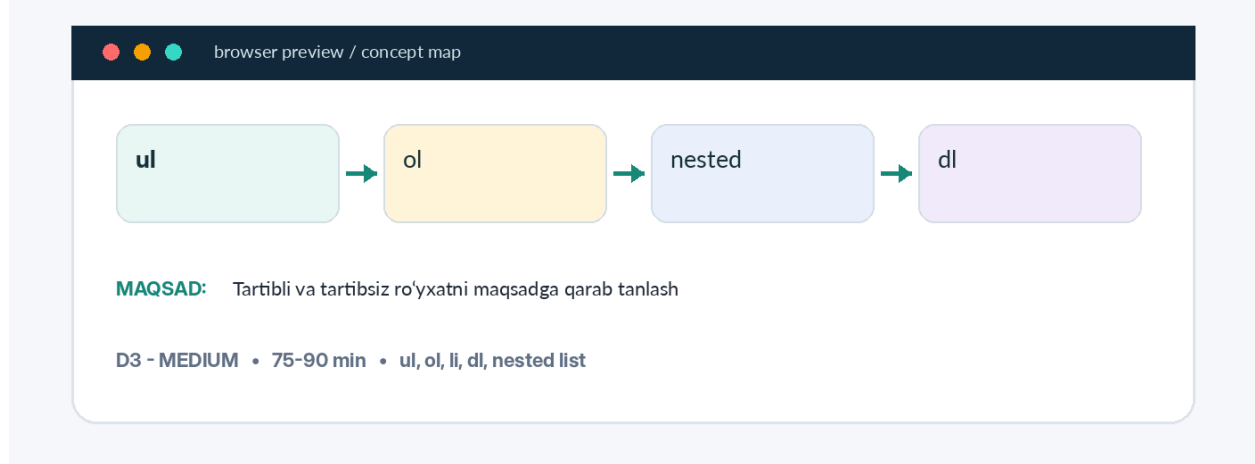
Nested

```
<ul>
  <li>Frontend
    <ul><li>HTML</li><li>CSS</li></ul>
  </li>
</ul>
```

Definition list

```
<dl>
  <dt>HTML</dt>
  <dd>Web-sahifa strukturası.</dd>
</dl>
```

CHAPTER 07 • Ro'yxatlar va menyu tuzilmalari



Vizual 7. Ro'yxatlar va menyu tuzilmalari - konsept xaritasi

KO'P UCHRAYDIGAN XATOLAR

- li ni ul/ol tashqarisida yozish.
- Ichki ro'yxatni li yopilgandan keyin joylashtirish.
- Oddiy paragrafni majburan ro'yxatga aylantirish.

Quick Check

1. Qachon ol ul dan ma'noliroq?
2. Ichki ro'yxat qaysi element ichida turishi kerak?
3. dl da dt va dd qanday rollarga ega?

MINI LOYIHA

- Kurs dasturini yarating: 3 modul ol ichida, har modulda mavzular ul ko'rinishida. Pastda 3 ta termin uchun dl qo'shing.

Challenges

Avval mustaqil bajaring. Kod ishlagach, semantika va accessibilityni ham tekshiring. Mumkin bo'lgan yechimlar kitobning eng oxiridagi PART III da.

061

D3

Uchta dasturlash yo'nalishidan iborat ul ro'yxat yarating.

062

D3

1 dan 4 gacha bosqichlardan iborat ol ro'yxat yarating.

063

ol ni 5-raqamdan boshlang.

D3

064

D3

Teskari sanaydigan ol ro'yxat yarating.

065

D3

Frontend ichida HTML va CSS bo'lgan ichma-ich ro'yxat tuzing.

066

D3

HTML va CSS uchun dl/dt/dd ta'rif ro'yxati yarating.

067

D3

Nav menyusini ul va linklar yordamida yozing.

068

D3

Xarid ro'yxatini checkboxsiz oddiy ul ko'rinishida yarating.

069

D3

3 bo'limli kurs dasturini ol va ichma-ich ul bilan yarating.

070

D3

Retsept uchun ingredientlarni ul, tayyorlash bosqichlarini ol bilan yozing.

CHAPTER CHECKPOINT • 07 YAKUNLANDI**MEN ENDI:**

- ✓ ul/olni mazmun tartibiga qarab tanlay olaman.
- ✓ Ichma-ich ro'yxatlarni to'g'ri nesting qilaman.
- ✓ dl/dt/dd bilan termin-izoh juftliklarini tuza olaman.

STRETCH • JAVOBSIZ

3 darajali kurs roadmapini ro'yxatlar bilan tuzing va pastida 5 terminli mini glossary qo'shing.

Jadvallar: struktura, birlashtirish va accessibility

DARAJA VA YO'NALISH

- D3 - MEDIUM+ • 90-110 min
- Fokus: table, caption, thead, scope, colspan, rowspan

Bu chapterda siz

- Jadvalni semantik qismlarga bo'lish
- Ustun va qator sarlavhalarini scope bilan belgilash
- colspan/rowspan ni ehtiyotkor ishlatish

Tushuntirish

Jadval qachon kerak?

Jadval faqat qator va ustunlar orasida mantiqiy bog'lanish bo'lgan ma'lumot uchun ishlatiladi: dars jadvali, narxlar, natijalar. Layout qilish uchun table ishlatish zamonaviy webda noto'g'ri yondashuv; sahifa layouti CSS bilan quriladi.

Asosiy tuzilma

<table> konteyner, <tr> qator, <th> sarlavha katagi, <td> ma'lumot katagi. <caption> jadvalning maqsadini ko'rinadigan tarzda tushuntiradi. Katta jadvalda <thead>, <tbody>, <tfoot> mazmunni mantiqiy guruhlaydi.

scope va accessibility

<th scope="col"> ustun, <th scope="row"> qator sarlavhasini bildiradi. Ekran o'quvchi katta jadvalda katak qaysi sarlavha bilan bog'liq ekanini shu ma'lumot yordamida yaxshiroq tushunadi. Murakkab jadvalni keragidan ortiq birlashtirish accessibilityni qiyinlashtiradi.

colspan va rowspan

colspan katakni bir necha ustun, rowspan bir necha qator bo'ylab kengaytiradi. Birlashganda har qatordagi mantiqiy ustun soni mos kelishini tekshiring. Avval oddiy jadval tuzing, keyin zarur bo'lsa birlashtiring.

Example Code

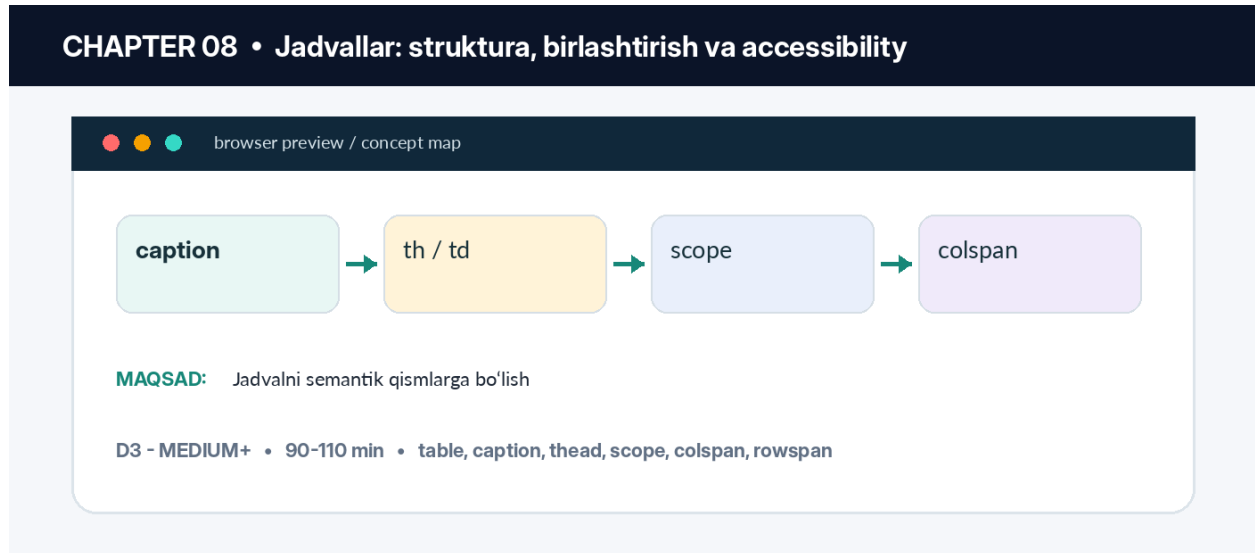
To'liq jadval

```
<table>
  <caption>Kurs natijalari</caption>
  <thead><tr><th scope="col">Talaba</th><th scope="col">Ball</th></tr></thead>
  <tbody><tr><th scope="row">Ali</th><td>88</td></tr></tbody>
  <tfoot><tr><td colspan="2">1 ta natija</td></tr></tfoot>
</table>
```

Rowspan

```
<tr><th rowspan="2" scope="rowgroup">HTML</th><td>A guruh</td></tr>
<tr><td>B guruh</td></tr>
```

CHAPTER 08 • Jadvallar: struktura, birlashtirish va accessibility



Vizual 8. Jadvallar: struktura, birlashtirish va accessibility - konsept xaritasi

KO'P UCHRAYDIGAN XATOLAR

- Jadvalni sahifa layouti uchun ishlatish.
- th o'rniga hamma joyda td yozish.
- colspan/rowspan dan keyin kataklar sonini tekshirmaslik.

Quick Check

1. caption foydalanuvchiga nima beradi?
2. scope="row" qayerda ishlatiladi?
3. colspan="2" nimani anglatadi?

MINI LOYIHA

- 4 mahsulotli narxlar jadvali tuzing: caption, thead, tbody, tfoot, scope va kamida bitta colspan ishlatib. Jadvalni ekran o'quvchi nuqtayi nazaridan tekshiring.

Challenges

Avval mustaqil bajaring. Kod ishlagach, semantika va accessibilityni ham tekshiring. Mumkin bo'lgan yechimlar kitobning eng oxiridagi PART III da.

071

D3

2 ustun, 3 qatorli oddiy jadval yarating.

072

D3

Jadvalga caption qo'shing.

073

D3

Ustun sarlavhalariga scope="col" qo'shing.

074

D3

thead va tbody ishlatilgan jadval tuzing.

075

D3

tfoot da umumiy qiymat ko'rsatilgan jadval yarating.

076

D3

Ikki guruh natijalari jadvalini yarating: caption, ustun sarlavhalari scope="col" va guruh nomi uchun colspan ishlatish.

077

D3

Fan nomi ikki qatorga tegishli bo'lgan jadval yarating: rowspan va ustun sarlavhalarida scope ishlatish.

078

D3

Kun, Vaqt, Fan, Xona ustunlari bilan 4 ta darslik jadval yarating. thead va tbody ishlatish.

079

D3

Talaba natijalari jadvali tuzing: har talaba ismi scope="row", fanlar scope="col" bo'lsin.

080

D3

To'liq narxlar jadvali yarating: caption, thead, tbody, tfoot, kamida 4 mahsulot va tfootda colspan ishlatish.

CHAPTER CHECKPOINT • 08 YAKUNLANDI**MEN ENDI:**

- ✓ caption, th, td, thead/tbody/tfoot vazifasini bilaman.
- ✓ scope yordamida jadval accessibilityini yaxshilay olaman.
- ✓ colspan/rowspan ishlatilganda mantiqiy ustunlarni tekshiraman.

STRETCH • JAVOBSIZ

Turnir natijalari uchun guruhlar, jamoalar, ochkolar va jami qatori bo'lgan accessible scoreboard yarating.

Formalar: input, label va foydalanuvchi ma'lumoti

DARAJA VA YO'NALISH

- D4 - INTERMEDIATE • 100-120 min
- Fokus: form, label, input, select, textarea, validation

Bu chapterda siz

- Formaning action/method/name mexanizmini tushunish
- Inputlarni label bilan to'g'ri bog'lash
- Brauzer validatsiya atributlarini maqsadga muvofiq ishlatish

Tushuntirish

Forma qanday ishlaydi?

<form> foydalanuvchi kiritgan qiymatlarni yuborish chegarasini belgilaydi. action ma'lumot qayerga yuborilishini, method esa odatda GET yoki POST usulini ko'rsatadi. HTML forma serverni o'zi yaratmaydi; real qabul qilish va saqlash backend vazifasidir.

name, id va label

Input yuborilganda uning name qiymati kalit sifatida ishlatiladi. id hujjat ichida elementni noyob aniqlaydi. <label for="email"> ning for qiymati input id si bilan mos bo'lsa, label bosilganda input fokus oladi va ekran o'quvchi maydon nomini aniq aytadi.

Input turlari

type="email" email formatiga, type="number" sonli boshqaruvga, type="date" sana tanlashga yordam beradi. Radio variantlardan bittasini tanlash uchun bir xil name ishlatiladi. Checkbox bir nechta mustaqil tanlovga mos. select tayyor variantlar, textarea esa ko'p satrli matn uchun ishlatiladi.

Brauzer validatsiyasi

required, minlength, maxlength, min, max, pattern kabi atributlar xatoni erta aniqlashga yordam beradi. Ammo ular xavfsizlik chegarasi emas: server tomonida ma'lumot yana tekshirilishi shart. Placeholder label o'rnini bosa olmaydi; u faqat qo'shimcha misol yoki hint sifatida ishlatiladi.

Example Code

Label + input

```
<label for="email">Email</label>
<input id="email" name="email" type="email" autocomplete="email" required>
```

Radio guruh

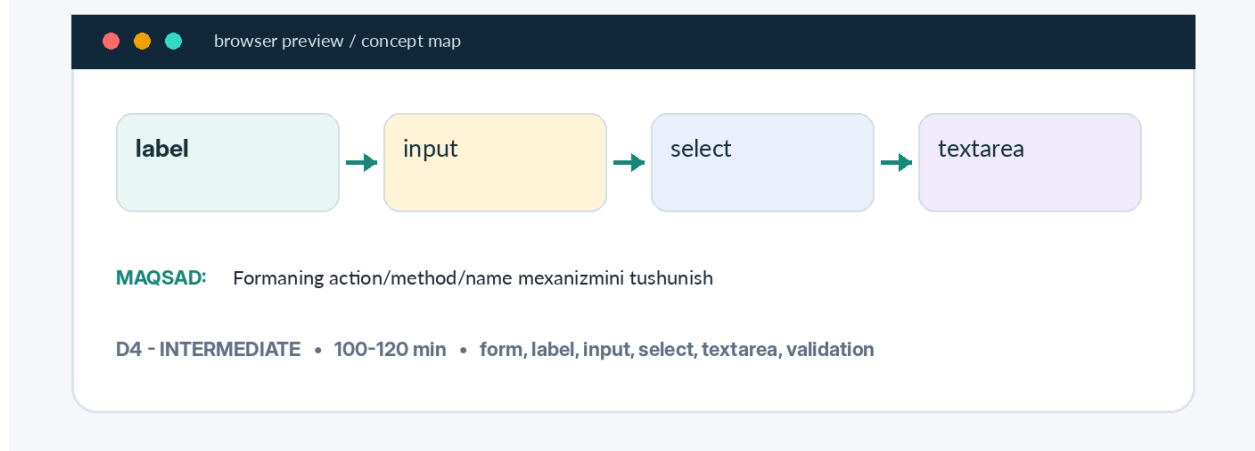
```
<fieldset>
  <legend>Rol</legend>
  <label><input type="radio" name="role" value="student"> Talaba</label>
  <label><input type="radio" name="role" value="teacher"> O'qituvchi</label>
</fieldset>
```

Select va textarea

```
<label for="city">Shahar</label>
<select id="city" name="city"><option>Toshkent</option><option>Buxoro</option></select>
```

```
<label for="message">Xabar</label>
<textarea id="message" name="message" rows="5"></textarea>
```

CHAPTER 09 • Formalar: input, label va foydalanuvchi ma'lumoti



Vizual 9. Formalar: input, label va foydalanuvchi ma'lumoti - konsept xaritasi

KO'P UCHRAYDIGAN XATOLAR

- inputda name atributini unutish.
- Placeholderni label o'rniga ishlatish.
- Maxfiy ma'lumotni GET query string orqali yuborish.

Quick Check

1. id va name form inputida qanday farq qiladi?
2. Nega radio variantlar bir xil name ishlatadi?
3. Client-side validation server validationni almashtiradimi?

MINI LOYIHA

- Kontakt formasi yarating: ism, email, mavzu select, xabar textarea, rozilik checkbox va submit tugmasi. Har maydon label bilan bog'langan va zarur maydonlar required bo'lsin.

Challenges

Avval mustaqil bajaring. Kod ishlagach, semantika va accessibilityni ham tekshiring. Mumkin bo'lgan yechimlar kitobning eng oxiridagi PART III da.

081

D4

Ism uchun label va text input yarating.

082

D4

Email uchun required input yarating.

083

D4

Parol maydoni yarating.

084

D4

Yosh uchun 1–120 oralig'idagi number input yarating.

085

D4

Ikki radio variant: Talaba va O'qituvchi.

086

D4

HTML va CSS qiziqishlari uchun checkboxlar yarating.

087

D4

Shahar tanlash uchun select yozing.

088

D4

Xabar uchun textarea yarating.

089

D4

To'liq kontakt forma yarating: ism, email, mavzu select, xabar textarea, label, required va submit tugmasi.

090

D4

Ro'yxatdan o'tish formasi yarating: ism, familiya, email, tug'ilgan sana, parol, rol radio, qiziqish checkbox, shahar select va submit tugmasi.

CHAPTER CHECKPOINT • 09 YAKUNLANDI**MEN ENDI:**

- ✓ label, id va name rollarini farqlay olaman.
- ✓ Input turini ma'lumot turiga mos tanlayman.
- ✓ Client-side validation server tekshiruvini almashtirmasligini tushunaman.

STRETCH • JAVOBSIZ

Stipendiya arizasi uchun 8+ maydonli accessible forma tuzing; radio, checkbox, select va textarea ishlatib.

Murakkab formalar, fieldset va umumiy atributlar

DARAJA VA YO'NALISH

- D4 - INTERMEDIATE • 110-130 min
- Fokus: fieldset, legend, datalist, file, pattern, data-*

Bu chapterda siz

- Katta formani mantiqiy fieldsetlarga bo'lish
- Datalist, file va pattern kabi imkoniyatlardan foydalanish
- id, class, data-* va autocomplete atributlarini maqsadga mos qo'llash

Tushuntirish

fieldset va legend

Murakkab formadagi bir-biriga tegishli boshqaruvlar `<fieldset>` ichida guruhlanadi, `<legend>` esa guruh nomini beradi. Bu ayniqsa radio/checkbox guruhlarida foydalanuvchi va ekran o'quvchi uchun kontekstni saqlaydi.

Datalist va file

`<datalist>` inputga tavsiya variantlarini beradi, lekin selectdan farqli ravishda foydalanuvchi boshqa qiymat ham yozishi mumkin. `type="file"` fayl tanlashga xizmat qiladi; `accept` atributi kutilayotgan formatlarni ko'rsatadi, ammo serverda fayl turi va hajmini baribir tekshirish kerak.

Pattern va autocomplete

`pattern` regular expression asosida matn formatiga cheklov beradi. `title` yoki yordamchi matn foydalanuvchiga talabni tushuntirishi kerak. `autocomplete="name", "email", "tel"` kabi tokenlar brauzerga ma'lumotni qulay to'ldirishga yordam beradi.

Umumiy atributlar

`id` noyob identifikator, `class` bir nechta elementni guruhlash uchun. `data-*` loyiha uchun maxsus metadata saqlaydi, masalan `data-step="2"`. `hidden` input foydalanuvchiga ko'rinmaydigan forma qiymatini yuborishi mumkin, lekin maxfiy ma'lumotni `hidden` ichiga ishonib saqlash xavfsiz emas.

Example Code

Fieldset

```
<fieldset>
  <legend>Shaxsiy ma'lumotlar</legend>
  <label for="name">Ism</label>
  <input id="name" name="name" autocomplete="name" required>
</fieldset>
```

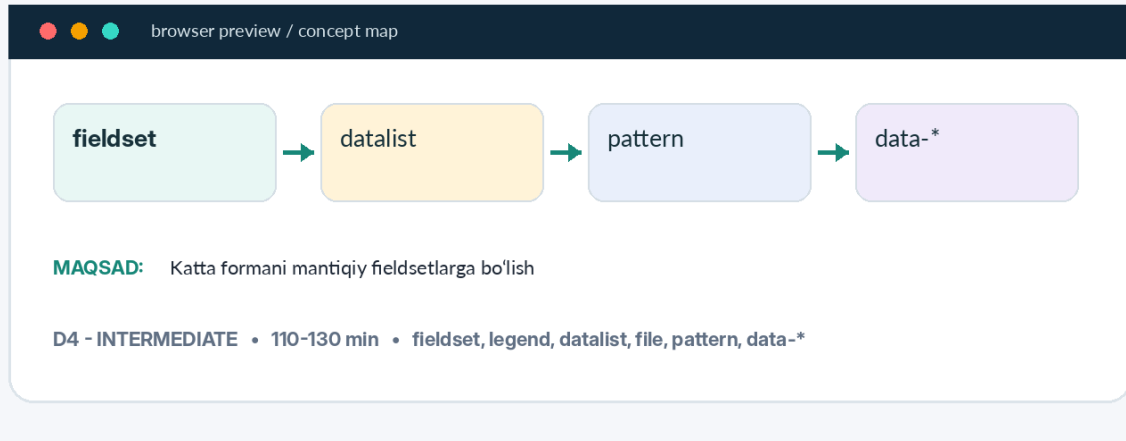
Datalist

```
<label for="city">Shahar</label>
<input id="city" name="city" list="cities">
<datalist id="cities">
  <option value="Toshkent">
  <option value="Buxoro">
</datalist>
```

Pattern

```
<label for="username">Login</label>
<input id="username" name="username" pattern="[A-Za-z0-9_]{4,16}" title="4-16 ta lotin harfi,
raqam yoki _">
```

CHAPTER 10 • Murakkab formalar, fieldset va umumiy atributlar



Vizual 10. Murakkab formalar, fieldset va umumiy atributlar - konsept xaritasi

KO'P UCHRAYDIGAN XATOLAR

- Radio guruhini legendsiz tashlab qo'yish.
- accept atributini xavfsizlik tekshiruvi deb hisoblash.
- Bir xil id ni bir nechta elementga berish.

Quick Check

1. fieldset va legend nima uchun birga ishlatiladi?
2. datalist selectdan qanday farq qiladi?
3. data-* ichidagi qiymat foydalanuvchidan yashirinmi?

MINI LOYIHA

- Ikki fieldsetli ro'yxatdan o'tish formasi yarating: shaxsiy ma'lumotlar va qiziqishlar. Datalist, file upload, pattern, autocomplete va hidden form_version ishlating.

Challenges

Avval mustaqil bajaring. Kod ishlagach, semantika va accessibilityni ham tekshiring. Mumkin bo'lgan yechimlar kitobning eng oxiridagi PART III da.

091

D4

Shaxsiy ma'lumotlar uchun fieldset/legend yarating; ism va familiya inputlari label bilan bog'lansin.

092

D4

Email va telefon maydonlarini autocomplete va tushunarli placeholder bilan yozing.

093

D4

Talaba/O'qituvchi/Dasturchi variantlari uchun fieldset va legend ichida radio guruh yarating.

094

D4

HTML, CSS, JavaScript qiziqishlari uchun label bilan bog'langan checkbox guruh yarating.

095

D4

Shahar kiritish uchun input + datalist yarating; Toshkent, Buxoro, Navoiy variantlari bo'lsin.

096

D4

Profil rasmi yuklash uchun file input yarating; faqat PNG va JPEG tavsiya qilinsin.

097

D4

Username uchun 4-16 belgilik lotin harfi, raqam yoki underscore qabul qiladigan pattern yozing va foydalanuvchiga talabni tushuntiring.

098

D4

Ikki bosqichli forma markup yarating: bo'limlarda class va data-step, shuningdek hidden form_version qiymati bo'lsin.

099

D4

So'rovnoma yarating: ikki fieldset, bitta select, textarea, required maydonlar, submit va reset tugmalari.

100

D4

Murakkab ro'yxatdan o'tish formasi yarating: ikki fieldset, autocomplete, radio, checkbox, datalist, file upload, pattern va hidden versionni birlashtiring.

CHAPTER CHECKPOINT • 10 YAKUNLANDI**MEN ENDI:**

- ✓ fieldset/legend bilan katta formani guruhlay olaman.
- ✓ datalist, file, pattern va autocomplete ishlata olaman.
- ✓ data-* va hidden qiymatlarning xavfsizlik chegarasini tushunaman.

STRETCH • JAVOBSIZ

2 bosqichli onboarding formasi yarating: profil + aloqa; file upload, datalist, pattern va version hidden qiymatini qo'shing.

Semantic HTML va sahifa arxitekturasini

DARAJA VA YO'NALISH

- D4 - INTERMEDIATE+ • 100-120 min
- Fokus: header, nav, main, section, article, aside, footer

Bu chapterda siz

- Sahifani ma'noli yirik hududlarga ajratish
- section va article farqini amalda qo'llash
- Div-soup markupni semantik strukturaga refaktor qilish

Tushuntirish

Nega semantic HTML?

Semantic element nomi uning vazifasini bildiradi. Bu kodni jamoada o'qishni, qidiruv tizimlari va yordamchi texnologiyalar bilan ishlashni yaxshilaydi. Har joyda div ishlash texnik jihatdan mumkin, lekin ma'no yo'qoladi.

Landmark elementlar

<header> kirish sarlavha hududi, <nav> muhim navigatsiya, <main> sahifaning asosiy noyob mazmuni, <footer> yakuniy hudud. Odatda sahifada bitta ko'rinadigan main landmark bo'ladi. header/footer faqat body darajasida emas, article/section ichida ham ishlatilishi mumkin.

section, article va aside

<article> mustaqil tarqatish yoki alohida o'qish mumkin bo'lgan kontent: blog posti, yangilik, forum posti. <section> tematik guruh; odatda sarlavhaga ega. <aside> asosiy mazmunga bog'liq, ammo chetga chiqarish mumkin bo'lgan qo'shimcha kontent.

details, time va refaktor

<details>/<summary> oddiy ochilib-yopiladigan disclosure beradi. <time datetime="2026-09-12"> kabi markup sana/vaqtini mashina o'qiydigan ko'rinishda ifodalaydi. Refaktorda avval kontent rolini aniqlang, keyin divni ma'noli elementga almashtiring; shunchaki barcha divni sectionga aylantirish ham xato.

Example Code

Sahifa skeleti

```
<header>...</header>
<nav aria-label="Asosiy navigatsiya">...</nav>
<main>
  <article>...</article>
  <aside>...</aside>
</main>
<footer>...</footer>
```

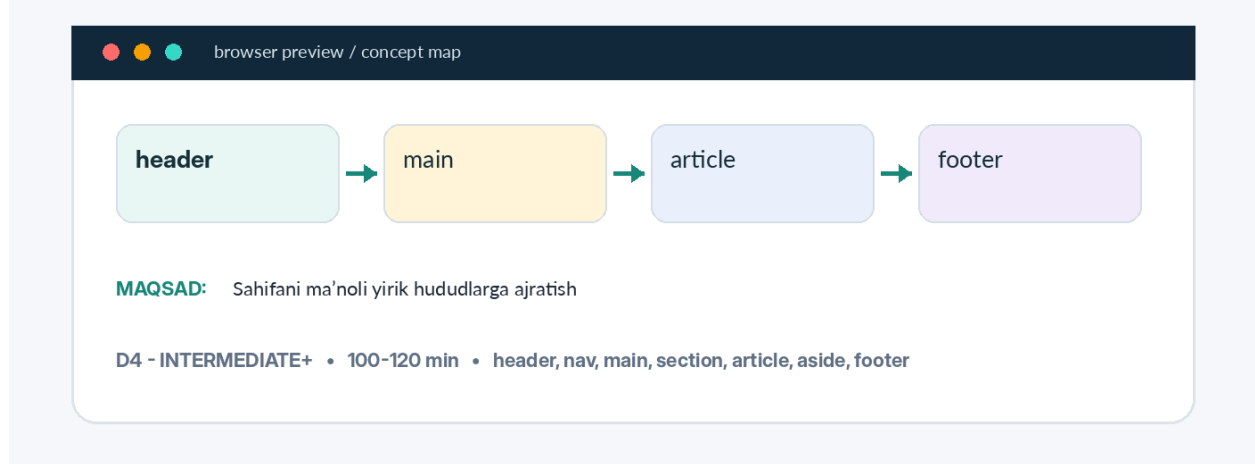
Article

```
<article>
  <header><h2>Yangi dars</h2><time datetime="2026-09-12">12-sentabr</time></header>
  <p>Maqola matni...</p>
  <footer><p>Muallif: Ali</p></footer>
</article>
```

FAQ

```
<details>
  <summary>HTML qiyinmi?</summary>
  <p>Asoslari amaliyot bilan tez o'zlashtiriladi.</p>
</details>
```

CHAPTER 11 • Semantic HTML va sahifa arxitekturası



Vizual 11. Semantic HTML va sahifa arxitekturası - konsept xaritasi

KO'P UCHRAYDIGAN XATOLAR

- Hamma divni avtomatik section bilan almashtirish.
- section ni sarlavhasiz ko'paytirish.
- Bir sahifada maqsadsiz bir nechta main landmark yaratish.

Quick Check

1. article bilan sectionning amaliy farqi nima?
2. aside asosiy maqolaning bir qismi bo'lishi mumkinmi?
3. Semantic teg tanlashning uch foydasini ayting.

MINI LOYIHA

- Oddiy divlardan iborat blog markup yozing, keyin uni header/nav/main/article/aside/footer yordamida refaktor qiling. Oldin va keyin variantni taqqoslang.

Challenges

Avval mustaqil bajaring. Kod ishlagach, semantika va accessibilityni ham tekshiring. Mumkin bo'lgan yechimlar kitobning eng oxiridagi PART III da.

101

D4

header ichida h1 va qisqa slogan yozing.

102

D4

nav ichida uchta havola yarating.

103

main elementiga maqola joylashtiring.

D4

104

D4

main ichida uchta ma'noli section yarating: "Kurs haqida", "Dastur", "Natijalar". Har section sarlavhaga ega bo'lsin.

105

D4

To'liq article yozing: header, h2, muallif, time, ikki paragraf va article footer.

106

D4

aside ichida "O'xshash mavzular" sarlavhasi va uchta tegishli havola yarating.

107

D4

Sayt footerini tuzing: copyright, email, telefon va maxfiylik siyosati linki.

108

D4

details/summary yordamida 3 ta FAQ savol-javobi yarating.

109

D4

Semantic kurs bosh sahifasi skeletini yozing: header, nav, main, kamida 3 section va footer.

110

D4

Divlardan iborat blog tuzilmasini semantic header/nav/main/article/aside/footer elementlariga refaktor qiling.

CHAPTER CHECKPOINT • 11 YAKUNLANDI

MEN ENDI:

- ✓ header/nav/main/article/section/aside/footer ni ma'noga mos tanlayman.
- ✓ Article va section farqini amalda tushunaman.
- ✓ Div-soup markupni semantic tuzilishga refaktor qila olaman.

STRETCH • JAVOBSIZ

Faqat divlardan tuzilgan yangilik sahifasini semantic HTMLga aylantiring va nima uchun o'zgartirganingizni 5 bandda yozing.

Audio, video, source va subtitrlar

DARAJA VA YO'NALISH

- D5 - ADVANCED • 100-120 min
- Fokus: audio, video, source, track, poster, transcript

Bu chapterda siz

- Audio/video elementlarini fallback bilan yozish
- Bir nechta media formatini source orqali taklif qilish
- Subtitr va transcript bilan media accessibilityni yaxshilash

Tushuntirish

Native media

`<audio>` va `<video>` zamonaviy brauzerda mediasiz pluginlarsiz ishlaydi. controls foydalanuvchiga play, pause va ovoz boshqaruvini beradi. controlsni olib tashlashdan oldin o'zingiz custom boshqaruv yaratishingiz va accessibilityni saqlashingiz kerak.

source va format

Element ichida bir nechta `<source>` berilsa brauzer qo'llab-quvvatlaydigan birinchi mos formatni tanlaydi. type atributi brauzerga faylni to'liq yuklamasdan format haqida signal beradi. Audio uchun MP3/OGG, video uchun MP4/WebM ko'p uchraydi.

Poster, preload va autoplay

poster video boshlanishidan oldingi rasmni ko'rsatadi. preload brauzerga oldindan yuklash bo'yicha hint beradi. Ovozli autoplay ko'pincha brauzer tomonidan bloklanadi va foydalanuvchiga noqulay; undan qochish ma'qul. loop faqat mazmun bunga mos bo'lsa ishlatiladi.

Subtitr va transcript

`<track kind="subtitles">` VTT subtitr faylini ulyadi. srclang va label tilni ko'rsatadi. Eshitishda qiyinchiligi bo'lgan foydalanuvchilar va qidiruv/tez o'qish uchun matn transcript ham foydali. Muhim video faqat tasvir orqali ma'no bersa audio description ehtiyoji ham ko'rib chiqiladi.

Example Code

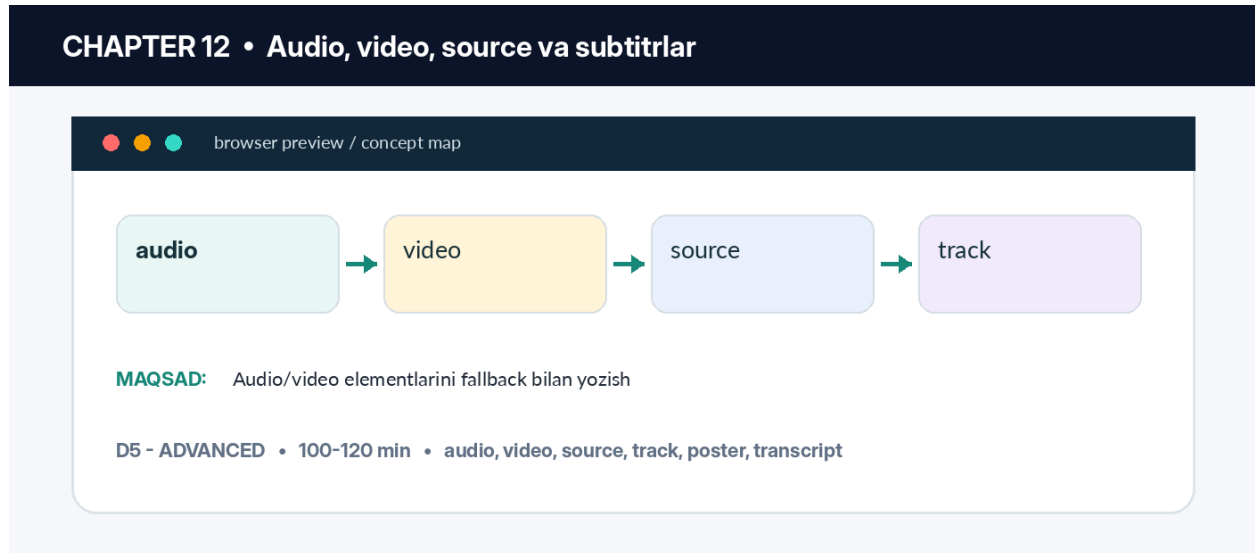
Audio source

```
<audio controls>
  <source src="media/lesson.mp3" type="audio/mpeg">
  <source src="media/lesson.ogg" type="audio/ogg">
  Audio ochildi.
</audio>
```

Video + track

```
<video controls width="720" poster="images/poster.jpg">
  <source src="media/lesson.webm" type="video/webm">
  <source src="media/lesson.mp4" type="video/mp4">
  <track src="media/lesson-uz.vtt" kind="subtitles" srclang="uz" label="0'zbekcha" default>
</video>
```

CHAPTER 12 • Audio, video, source va subtitrlar



Vizual 12. Audio, video, source va subtitrlar - konsept xaritasi

KO'P UCHRAYDIGAN XATOLAR

- Muhim media boshqaruvini controls siz qoldirish.
- Subtitrsiz uzoq video berish.
- Ovozli autoplayni foydalanuvchi roziligisiz ishlatish.

Quick Check

1. source tartibi brauzerga qanday ta'sir qiladi?
2. trackdagi srclang nima?
3. Subtitr va transcript bir xil narsa emasligi nimada?

MINI LOYIHA

- Dars sahifasi yarating: audio, posterli video, MP4/WebM source, o'zbekcha VTT subtitle va video ostida matn transcript linki bo'lsin.

Challenges

Avval mustaqil bajaring. Kod ishlagach, semantika va accessibilityni ham tekshiring. Mumkin bo'lgan yechimlar kitobning eng oxiridagi PART III da.

111

D5

lesson.mp3 uchun controls li audio yarating.

112

D5

MP3 va OGG source li audio yozing.

113

D5

intro.mp4 uchun video element yarating.

114

D5

Videoga poster rasm qo'shing.

115

D5

MP4 va WebM source li video yozing.

116

D5

Videoga o'zbekcha subtitr track qo'shing.

117

D5

Audio elementga loop atributini qo'shing.

118

D5

Media dars sahifasi yarating: sarlavha, source li audio, posterli MP4/WebM video, o'zbekcha subtitle track va transcript linki.

CHAPTER CHECKPOINT • 12 YAKUNLANDI**MEN ENDI:**

- ✓ audio/video uchun source va fallback yozaman.
- ✓ poster, track va srclang bilan video tuza olaman.
- ✓ Subtitr va transcript farqini tushunaman.

STRETCH • JAVOBSIZ

Accessible "video dars" sahifasi tuzing: 2 format, poster, VTT subtitr, transcript va fallback matn.

iframe, object, embed va xavfsiz tashqi kontent

DARAJA VA YO'NALISH

- D5 - ADVANCED • 100-120 min
- Fokus: iframe, title, loading, sandbox, object fallback

Bu chapterda siz

- iframe ni title va lazy loading bilan to'g'ri ishlatish
- sandbox orqali tashqi kontent ruxsatlarini cheklash
- PDF va boshqa resurslarga fallback berish

Tushuntirish

Iframe nima?

<iframe> boshqa HTML hujjatni joriy sahifa ichida alohida browsing context sifatida ko'rsatadi. Ichki demo, xarita, video player yoki ishonchli tashqi vidjet uchun ishlatilishi mumkin. Har iframe uchun mazmunini tushuntiruvchi title accessibility uchun muhim.

Lazy loading va o'lcham

loading="lazy" ekranning ancha pastidagi iframe ni keyinroq yuklashga yordam beradi. width/height joy ajratadi, ammo responsive layout uchun keyinchalik CSS kerak bo'ladi. Og'ir iframe sahifa tezligiga sezilarli ta'sir qilishi mumkin.

Sandbox xavfsizligi

sandbox atributi iframe dagi skript, forma, popup, same-origin kabi imkoniyatlarni cheklaydi. Kerakli ruxsatni allow-forms, allow-scripts kabi token bilan alohida ochish mumkin. Ishonchsiz kontentga keragidan ortiq ruxsat berish xavfli; "hammasini ochib qo'yish" sandbox maqsadini yo'qqa chiqaradi.

object va embed

PDF kabi faylni <object> yoki <embed> bilan ko'rsatish mumkin. <object> ichida fallback HTML joylashtirish qulay: resurs ochildmasa foydalanuvchi oddiy link orqali faylni yuklab oladi. Muhim hujjat uchun har doim bevosita link ham berish yaxshi amaliyot.

Example Code

Lazy iframe

```
<iframe src="demo.html" title="HTML demo" loading="lazy" width="800" height="500"></iframe>
```

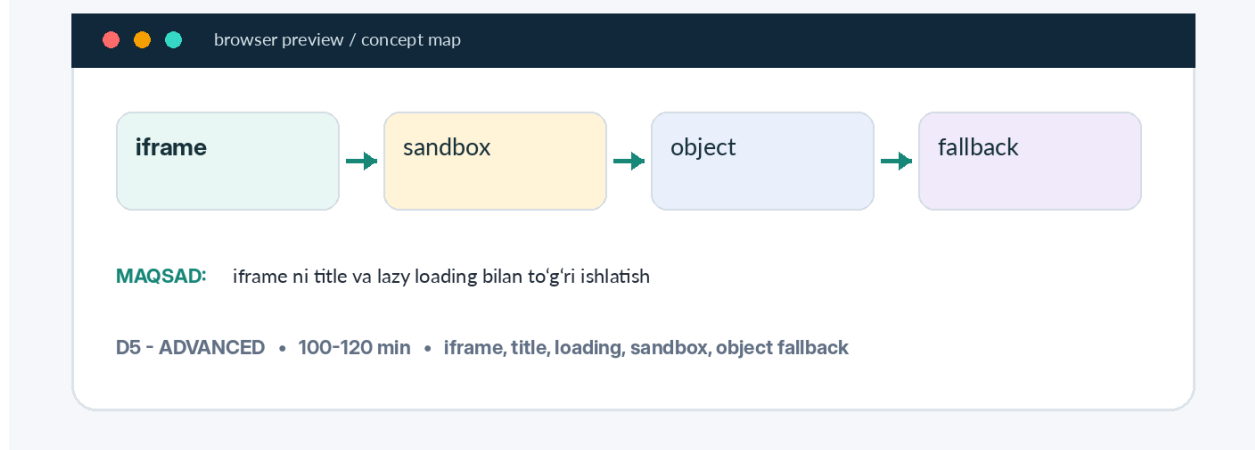
Sandbox

```
<iframe src="form-widget.html" title="Forma vidjeti" sandbox="allow-forms"></iframe>
```

Object fallback

```
<object data="files/guide.pdf" type="application/pdf" width="100%" height="700">
  <p>PDF ochildmadi. <a href="files/guide.pdf">Yuklab oling</a>.</p>
</object>
```

CHAPTER 13 • iframe, object, embed va xavfsiz tashqi kontent



Vizual 13. iframe, object, embed va xavfsiz tashqi kontent - konsept xaritasi

KO'P UCHRAYDIGAN XATOLAR

- iframe ga title bermaslik.
- Ishonchsiz iframe uchun ortiqcha sandbox ruxsatlarini ochish.
- PDF embedni fallback linksiz qoldirish.

Quick Check

1. iframe title kim uchun ayniqsa muhim?
2. sandbox bo'sh atribut holatida nimani anglatadi?
3. objectning embedga nisbatan foydali tomoni nima?

MINI LOYIHA

- "Resurslar" sahifasi yarating: ichki demo iframe lazy yuklansin, alohida sandbox iframe bo'lsin, PDF object orqali ko'rinsin va fallback download linki berilsin.

Challenges

Avval mustaqil bajaring. Kod ishlagach, semantika va accessibilityni ham tekshiring. Mumkin bo'lgan yechimlar kitobning eng oxiridagi PART III da.

119

D5

about.html ni iframe ichida ko'rsating.

120

D5

Iframe ga width va height qo'shing.

121

D5

Iframe ni lazy loading bilan yozing.

122

D5

Sandbox qilingan oddiy iframe yarating.

123

D5

Formalar uchun ruxsatli sandbox iframe misolini yozing.

124

D5

files/book.pdf ni embed orqali joylashtiring.

125

D5

PDF uchun object va fallback link yozing.

126

D5

Resurslar sahifasi yarating: lazy ichki iframe, sandbox qilingan forma iframe va PDF uchun object + fallback link.

CHAPTER CHECKPOINT • 13 YAKUNLANDI**MEN ENDI:**

- ✓ iframega title va loadingni to'g'ri beraman.
- ✓ sandbox ruxsatlarini minimal prinsipda tanlayman.
- ✓ object uchun fallback link beraman.

STRETCH • JAVOBSIZ

"Resurs markazi" sahifasida ichki demo, sandbox vidjet va PDF qo'llanmani xavfsiz joylashtiring.

head, metadata, SEO va social preview

DARAJA VA YO'NALISH

- D5 - ADVANCED • 100-120 min
- Fokus: title, description, canonical, robots, Open Graph

Bu chapterda siz

- Har sahifa uchun aniq title va description yozish
- Canonical/robots vazifasini tushunish
- Open Graph yordamida ijtimoiy ulashish previewini belgilash

Tushuntirish

Title va description

<title> sahifaning eng muhim metadata elementlaridan biri. U aniq, sahifaga xos va foydalanuvchiga tushunarli bo'lsin. meta description qidiruv tizimiga sahifa mazmuni haqida qisqa tavsif beradi; qidiruv tizimi uni natijada aynan ko'rsatishi kafolatlanmaydi.

Canonical va robots

<link rel="canonical"> bir xil yoki juda o'xshash kontentga ega URLlar orasida afzal manzilni ko'rsatishga yordam beradi. meta robots index/noindex va follow/nofollow kabi ko'rsatmalar beradi. Bu elementlar "SEO hiyla" emas; sayt arxitekturasini va indekslash niyatini aniq ifodalash vositalaridir.

Favicon va viewport

favicon brauzer tabida saytni tanitadi. viewport mobil ko'rinish uchun zarur asosiy metadata. charset ham head boshida erta berilgani ma'qul. Har sahifada takrorlanuvchi head template bo'lishi mumkin, ammo title/description mazmunga mos ravishda o'zgarishi kerak.

Open Graph

og:title, og:description, og:image va og:type kabi Open Graph teglar link ijtimoiy tarmoq yoki messengerda ulashilganda preview uchun metadata beradi. og:image odatda to'liq absolute URL bilan beriladi. Preview to'g'ri chiqishini platformaning debugger vositalari bilan tekshirish mumkin.

Example Code

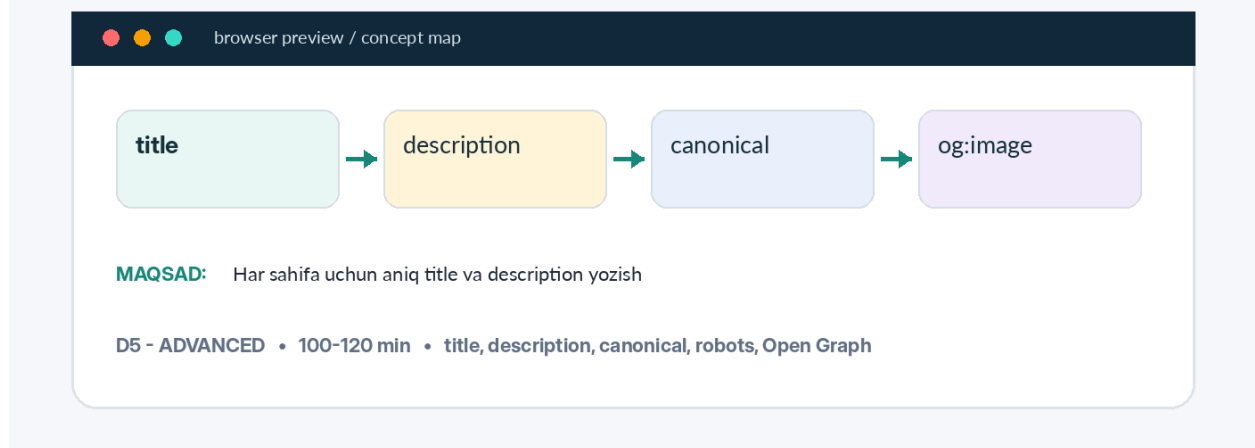
SEO head

```
<title>HTML kursi | Code Academy</title>
<meta name="description" content="HTMLni amaliy topshiriqlar bilan o'rganing.">
<link rel="canonical" href="https://example.com/html-course">
```

Social metadata

```
<meta property="og:title" content="HTML kursi">
<meta property="og:description" content="150 ta amaliy topshiriq">
<meta property="og:image" content="https://example.com/images/cover.jpg">
<meta property="og:type" content="website">
```

CHAPTER 14 • head, metadata, SEO va social preview



Vizual 14. head, metadata, SEO va social preview - konsept xaritasi

KO'P UCHRAYDIGAN XATOLAR

- Barcha sahifada bir xil title va description ishlatish.
- Canonicalga noto'g'ri yoki boshqa sahifa URLini berish.
- og:image uchun lokal relative manzil bilan cheklanib qolish.

Quick Check

1. Canonical qachon foydali?
2. robots noindex nimani bildiradi?
3. OG metadata qidiruv rankingini to'g'ridan-to'g'ri kafolatlaydimi?

MINI LOYIHA

- Kurs landing page uchun to'liq head yarating: charset, viewport, title, description, robots, canonical, favicon va to'rtta asosiy OG metadata.

Challenges

Avval mustaqil bajaring. Kod ishlagach, semantika va accessibilityni ham tekshiring. Mumkin bo'lgan yechimlar kitobning eng oxiridagi PART III da.

127

D5

"HTML kursi" title va description meta yozing.

128

D5

Viewport meta tegini yozing.

129

D5

favicon uchun link yozing.

130

D5

Canonical URL kiriting.

131

D5

Sahifani indekslashga ruxsat beruvchi robots meta yozing.

132

D5

OG title va OG description yozing.

133

D5

OG image va OG type qo'shing.

134

D5

SEO va social preview kiritilgan to'liq head yozing: charset, viewport, title, description, robots, canonical, favicon, og:title, og:description, og:image, og:type.

CHAPTER CHECKPOINT • 14 YAKUNLANDI**MEN ENDI:**

- ✓ Har sahifaga noyob title va description yozaman.
- ✓ Canonical va robotsning vazifasini tushunaman.
- ✓ Open Graph preview metadata tuza olaman.

STRETCH • JAVOBSIZ

Bir saytning Home, Kurs va Kontakt sahifalari uchun bir-biridan farqli, mazmunga mos 3 ta to'liq head yozing.

Accessibility: semantic, alt, label va klaviatura

DARAJA VA YO'NALISH

- D6 - HARD • 100-130 min
- Fokus: a11y, alt, label, scope, button, skip link

Bu chapterda siz

- Semantic HTMLni accessibility poydevori sifatida ishlatish
- Rasm, forma va jadvalni yordamchi texnologiyalar uchun tushunarli qilish
- Klaviatura va fokus nuqtayi nazaridan interaktiv elementlarni tanlash

Tushuntirish

Accessibility poydevori

Accessibility (a11y) saytni turli ko'rish, eshitish, harakat yoki kognitiv ehtiyojlarga ega foydalanuvchilar uchun ishlatish mumkin bo'lishini anglatadi. Birinchi qadam ko'pincha ARIA emas, to'g'ri native HTML: button uchun `<button>`, link uchun `<a>`, sarlavha uchun `h1-h6`.

Alt va ko'rinadigan matn

Mazmunli rasmning alt matni uning sahifadagi vazifasini ifodalashi kerak. Dekorativ rasm `alt=""` oladi. Rasmning o'zidagi barcha matnni ko'r-ko'rona takrorlash shart emas; kontekstga kerakli ma'lumotni bering. Muhim diagrammada yaqin atrofda batafsil matn tavsifi ham kerak bo'lishi mumkin.

Forma va jadval

Har form control aniq accessible namega ega bo'lishi kerak; eng oddiy usul label. Radio/checkbox guruhini fieldset/legend bilan guruhlash foydali. Jadvalda caption, th va scope munosabatni tushuntiradi. Placeholder label o'rnini bosmaydi.

Klaviatura, fokus va skip link

Native link va button klaviatura bilan ishlaydi; klik uchun div yaratish qo'shimcha keyboard/role/focus ishlarini talab qiladi. Sahifa boshidagi "Asosiy kontentga o'tish" skip link takroriy navigatsiyani tez chetlab o'tishga yordam beradi. Fokus tartibi DOM tartibiga mantiqan mos bo'lishi kerak.

Example Code

Skip link

```
<a href="#main">Asosiy kontentga o'tish</a>
<nav>...</nav>
<main id="main">...</main>
```

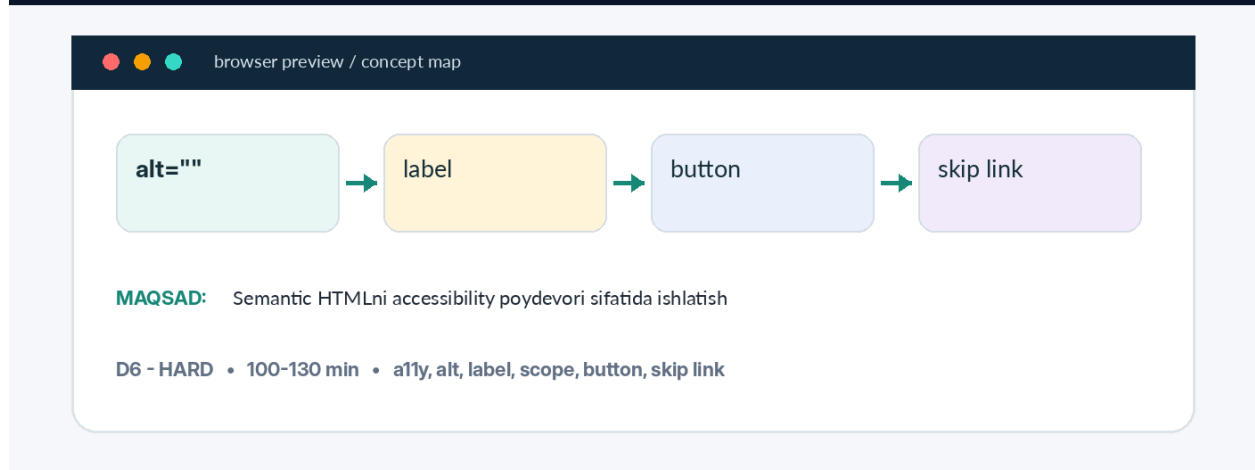
Accessible forma

```
<fieldset>
  <legend>Aloqa usuli</legend>
  <label><input type="radio" name="contact" value="email"> Email</label>
  <label><input type="radio" name="contact" value="phone"> Telefon</label>
</fieldset>
```

Native button

```
<button type="button">Menyuni ochish</button>
```

CHAPTER 15 • Accessibility: semantic, alt, label va klaviatura



Vizual 15. Accessibility: semantic, alt, label va klaviatura - konsept xaritasi

KO'P UCHRAYDIGAN XATOLAR

- Klik qilinadigan divni button o'rniga ishlatish.
- Placeholderni label sifatida qabul qilish.
- Heading darajasini faqat dizayn uchun buzish.

Quick Check

1. Dekorativ rasm uchun alt qanday bo'ladi?
2. Nega native button divdan yaxshiroq?
3. Skip linkning foydasi nima?

MINI LOYIHA

- Oldindan berilgan "yomon" sahifani audit qiling: alt, label, heading, table scope, click-div va skip-link muammolarini topib accessible variantga refaktor qiling.

Challenges

Avval mustaqil bajaring. Kod ishlagach, semantika va accessibilityni ham tekshiring. Mumkin bo'lgan yechimlar kitobning eng oxiridagi PART III da.

135

D5

Mazmunli profil rasmi uchun aniq alt yozing.

136

D5

Dekorativ bezak rasmiga bo'sh alt qo'ying.

137

D5

Email inputni label bilan to'g'ri bog'lang.

138

Ikki ustunli jadval sarlavhalariga scope qo'shing.

D5

139

D5

Klik qilinadigan div o'rniga haqiqiy button ishlating.

140

D5

Quyidagi tipik accessibility muammolarini bir sahifada tuzating: mazmunli/dekorativ rasm altlari, email label, jadval scope, klik qilinadigan div o'rniga button va skip link.

CHAPTER CHECKPOINT • 15 YAKUNLANDI

MEN ENDI:

- ✓ Alt, label, scope va native controls bilan a11yni yaxshilayman.
- ✓ Klaviatura fokusini va skip link ehtiyojini tushunaman.
- ✓ ARIA ni native HTML yetmaganda ishlatish kerakligini bilaman.

STRETCH • JAVOBSIZ

Oldingi loyihalardan bittasini tanlab 10 bandlik accessibility audit qiling va topilgan kamida 5 muammoni tuzating.

Integration, refaktor va debugging workshop

DARAJA VA YO'NALISH

- D6+ - HARD • 2-3 soat
- Fokus: integratsiya, validation, refactor, real page

Bu chapterda siz

- Bir nechta HTML ko'nikmasini bitta sahifada birlashtirish
- Noto'g'ri markupni sababini tushunib refaktor qilish
- Part II yakuniy loyihalariga mustaqil reja bilan tayyorlanish

Tushuntirish

Reja - koddan oldin

Murakkab sahifada avval kontent xaritasini yozing: sahifaning maqsadi, asosiy h1, navigatsiya, section/article lar, media, jadval yoki forma. Keyin fayl va papka strukturasi. Rejasiz markup yozish heading ierarxiyasi, takroriy id va chalkash nesting kabi xatolarni ko'paytiradi.

Validation va DevTools

HTML Validator sintaktik/standart xatolarni topadi, DevTools esa brauzer hosil qilgan real DOMni ko'rsatadi. Source kodi to'g'ri ko'ringani bilan parser noto'g'ri nestingni o'zi tuzatgan bo'lishi mumkin. Elements panelida elementlar qayerga ko'chganini tekshirish debuggingning muhim qismi.

Refaktor ketma-ketligi

Avval hujjat skeleti, keyin heading va landmarklar, so'ng link/path, rasm alt, jadval/form accessibility, eng oxirida metadata va kichik detallarni tekshiring. Bir vaqtda hamma narsani o'zgartirmang: kichik bosqichda test qilish regressiyani topishni osonlashtiradi.

Definition of done

Sahifa "brauzerda ochildi" degani tayyor degani emas. Yakuniy tekshiruv: title/description aniqmi, nav ishlaydimi, heading mantiqiyimi, rasm altlari to'g'rimi, form labelga egami, klaviatura bilan boshqariladimi, validator katta xato bermaydimi, mobil viewport mavjudmi. Part II da aynan shu checklist ishlatiladi.

Example Code

Debug checklist

1. DOCTYPE / lang / charset
2. Heading ierarxiyasi
3. Semantic landmarklar
4. Link va file path
5. Alt / label / scope
6. Validator + DevTools

Landing skeleti

```
<header>...</header>
<main id="main">
  <section id="hero">...</section>
  <section id="features">...</section>
  <section id="cta">...</section>
</main>
<footer>...</footer>
```

CHAPTER 16 • Integration, refaktor va debugging workshop



Vizual 16. Integration, refaktor va debugging workshop - konsept xaritasi

KO'P UCHRAYDIGAN XATOLAR

- Validator xabarini tushunmasdan kodni tasodifiy o'zgartirish.
- Bir xil id ni takrorlash.
- Semantik refaktorda dizayn classlarini ma'no bilan adashtirish.

Quick Check

1. DevTools DOM bilan View Source nimada farq qilishi mumkin?
2. Murakkab sahifani refaktor qilishda qaysi qatlamdan boshlash ma'qul?
3. "Definition of done" checklistiga kamida 5 band ayting.

MINI LOYIHA

- O'zingizning oldingi 100-topshiriq formangiz va 110-topshiriq semantic skeletingizni birlashtirib, validator/a11y checklistdan o'tkazing. Topilgan kamida 3 muammoni yozib, tuzating.

Challenges

Avval mustaqil bajaring. Kod ishlagach, semantika va accessibilityni ham tekshiring. Mumkin bo'lgan yechimlar kitobning eng oxiridagi PART III da.

141

D6

Ataylab buzilgan HTML hujjatni to'liq tuzating: doctype, lang, charset, title, body nesting, duplicate id, rasm alt, forma label va heading ierarxiyasi to'g'ri bo'lsin.

142

D6

"Div soup" blog sahifasini semantic HTMLga refaktor qiling: header, nav, main, article, figure, aside, footer va time ishlatib.

143

D6

Professional mahsulot sahifasi yarating: SEO head, semantic article, figure, narx, xususiyatlar ro'yxati, yetkazish jadvali va buyurtma button.

144

Kursga yozilish sahifasi yarating: kurs tavsifi, talablar ro'yxati, jadval va ikki fieldsetli accessible

D6

ariza formasi.

145
D6

Bir sahifali professional portfolio yarating: skip link, SEO head, nav, bio figure, skills ro'yxati, loyihalar jadvali, accessible kontakt forma va footer.

CHAPTER CHECKPOINT • 16 YAKUNLANDI

MEN ENDI:

- ✓ Murakkab sahifani koddan oldin rejalayman.
- ✓ Validator va DevToolsni birgalikda ishlataman.
- ✓ Definition of done bilan yakuniy QA qila olaman.

STRETCH • JAVOBSIZ

146-150 capstonelardan bittasiga 1 sahifalik loyiha briefi: fayl daraxti, kontent xaritasi, risklar va QA checklist tuzing.

PART II

YAKUNIY LOYIHALAR

146-150 • reja tuzing, fayllarni yarating, validator va checklist bilan tekshiring

Capstone qoidasi

Bu bo'limda tayyor yo'l ko'rsatilmaydi. Har loyiha oldingi chapterlardagi ko'nikmalarni birlashtiradi. Avval fayl strukturasi va kontent xaritasini rejalang. Har sahifani alohida tekshiring. Yakuniy kodlarning to'liq mumkin bo'lgan yechimlari kitobning eng oxirida berilgan.

YAKUNIY TEKSHIRUV

- DOCTYPE, lang, charset, viewport va noyob title/description
- Ishlaydigan navigatsiya va mantiqiy heading ierarxiyasi
- Semantic landmarklar va toza nesting
- Rasmlarda to'g'ri alt, form inputlarida label
- Jadvalda caption/th/scope; klaviaturada native interaktiv elementlar
- HTML validator va DevTools bilan yakuniy tekshiruv

CHALLENGE 146

Shaxsiy portfolio

CAPSTONE

- index.html, about.html, projects.html va contact.html dan iborat 4 sahifali portfolio yarating. Barcha sahifalarda bir xil navigatsiya va footer bo'lsin. Har sahifada noyob title/description, to'g'ri heading ierarxiyasi va semantic HTML ishlating. About sahifada figure va skills ro'yxati, Projects sahifada kamida 3 loyiha, Contact sahifada label bilan bog'langan accessible forma bo'lsin.

Topshirishdan oldingi checklist

- ☐ Barcha talab qilingan fayllar mavjud
- ☐ Har sahifa valid HTML5 skeletga ega
- ☐ Navigatsiyada broken link yo'q
- ☐ Har sahifada title va description mazmunga mos
- ☐ Heading ierarxiyasi mantiqiy
- ☐ Rasm altlari maqsadga mos
- ☐ Forma label va fieldset/legend bilan tushunarli
- ☐ Semantic elementlar ma'noga mos
- ☐ Klaviatura bilan asosiy harakatlar bajariladi
- ☐ Validator katta markup xatosi ko'rsatmaydi

LOYIHA REJASI • CHALLENGE 146

Fayl daraxti	_____
Asosiy maqsad / h1	_____
A11y + SEO fokus	_____
Yakuniy QA natijasi	Validator: _____ A11y: _____

CHALLENGE 147

Universitet sayti

CAPSTONE

- index.html, faculties.html, news.html va admission.html dan iborat 4 sahifali universitet sayti yarating. Fakultetlar section/article lar bilan, yangiliklar article/time bilan, qabul sahifasi esa dastur jadvali va to'liq ariza formasi bilan tuzilsin.

Topshirishdan oldingi checklist

- ☐ Barcha talab qilingan fayllar mavjud
- ☐ Har sahifa valid HTML5 skeletga ega
- ☐ Navigatsiyada broken link yo'q
- ☐ Har sahifada title va description mazmunga mos
- ☐ Heading ierarxiyasi mantiqiy
- ☐ Rasm altlari maqsadga mos
- ☐ Forma label va fieldset/legend bilan tushunarli
- ☐ Semantic elementlar ma'noga mos
- ☐ Klaviatura bilan asosiy harakatlar bajariladi
- ☐ Validator katta markup xatosi ko'rsatmaydi

LOYIHA REJASI • CHALLENGE 147

Fayl daraxti	_____
Asosiy maqsad / h1	_____
A11y + SEO fokus	_____
Yakuniy QA natijasi	Validator: _____ A11y: _____

CHALLENGE 148

Online do‘kon

CAPSTONE

- index.html, products.html, product.html va checkout.html dan iborat HTML prototip yarating. Katalogda kamida 3 mahsulot, product sahifada figure, narx, xususiyatlar va yetkazish jadvali, checkoutda ikki fieldsetli accessible forma bo‘lsin.

Topshirishdan oldingi checklist

- ☐ Barcha talab qilingan fayllar mavjud
- ☐ Har sahifa valid HTML5 skeletga ega
- ☐ Navigatsiyada broken link yo‘q
- ☐ Har sahifada title va description mazmunga mos
- ☐ Heading ierarxiyasi mantiqiy
- ☐ Rasm altlari maqsadga mos
- ☐ Forma label va fieldset/legend bilan tushunarli
- ☐ Semantic elementlar ma’noga mos
- ☐ Klaviatura bilan asosiy harakatlar bajariladi
- ☐ Validator katta markup xatosi ko‘rsatmaydi

LOYIHA REJASI • CHALLENGE 148

Fayl daraxti	_____
Asosiy maqsad / h1	_____
A11y + SEO fokus	_____
Yakuniy QA natijasi	Validator: _____ A11y: _____

CHALLENGE 149

Blog platformasi

CAPSTONE

- index.html, post.html, archive.html va about.html dan iborat blog prototip yarating. Bosh sahifada article ro'yxati va newsletter formasi, postda time/figure/aside, archive sahifada accessible jadval, aboutda muallif ma'lumoti bo'lsin.

Topshirishdan oldingi checklist

- ☐ Barcha talab qilingan fayllar mavjud
- ☐ Har sahifa valid HTML5 skeletga ega
- ☐ Navigatsiyada broken link yo'q
- ☐ Har sahifada title va description mazmunga mos
- ☐ Heading ierarxiyasi mantiqiy
- ☐ Rasm altlari maqsadga mos
- ☐ Forma label va fieldset/legend bilan tushunarli
- ☐ Semantic elementlar ma'noga mos
- ☐ Klaviatura bilan asosiy harakatlar bajariladi
- ☐ Validator katta markup xatosi ko'rsatmaydi

LOYIHA REJASI • CHALLENGE 149

Fayl daraxti	_____
Asosiy maqsad / h1	_____
A11y + SEO fokus	_____
Yakuniy QA natijasi	Validator: _____ A11y: _____

CHALLENGE 150

FINAL - Nova Studio

CAPSTONE

- Kamida 5 sahifali professional website yarating: index.html, about.html, services.html, gallery.html, contact.html. Barcha sahifalarda ishlaydigan nav, semantic HTML, noyob SEO head, accessibility, responsive rasm manbalari, xizmatlar jadvali, galereya, accessible kontakt forma va bitta PDF resurs uchun object fallback ishlatish.

Topshirishdan oldingi checklist

- ☐ Barcha talab qilingan fayllar mavjud
- ☐ Har sahifa valid HTML5 skeletga ega
- ☐ Navigatsiyada broken link yo'q
- ☐ Har sahifada title va description mazmunga mos
- ☐ Heading ierarxiyasi mantiqiy
- ☐ Rasm altlari maqsadga mos
- ☐ Forma label va fieldset/legend bilan tushunarli
- ☐ Semantic elementlar ma'noga mos
- ☐ Klaviatura bilan asosiy harakatlar bajariladi
- ☐ Validator katta markup xatosi ko'rsatmaydi

LOYIHA REJASI • CHALLENGE 150

Fayl daraxti	_____
Asosiy maqsad / h1	_____
A11y + SEO fokus	_____
Yakuniy QA natijasi	Validator: _____ A11y: _____

APPENDIKSLAR

QO'LLANMA VA INDEX

Glossary • teglar indeksi • debugging checklist • starter assetlar • 30 kunlik reja • rubrika • rasmiy manbalar

HTML glossary

62 ta asosiy atama. Atamalar alifbo tartibida berilgan.

ATAMA	IZOH
Absolute URL	Protokol va domenni ham o'z ichiga olgan to'liq internet manzil.
Accessibility (a11y)	Saytni turli ehtiyoj va imkoniyatlarga ega foydalanuvchilar uchun ishlatish mumkin bo'lishi.
Anchor	Sahifadagi id ga # orqali bog'langan ichki havola nuqtasi.
ARIA	Native HTML yetarli bo'lmagan ayrim interfeyslarda accessibility semantikasi beruvchi atributlar to'plami.
Attribute	Elementga qo'shimcha ma'lumot beruvchi name/value xususiyat.
Autocomplete	Brauzerga forma maydonini foydalanuvchi ma'lumoti bilan qulay to'ldirishga yordam beruvchi atribut.
Block element	Odatda yangi qatordan boshlanib blok oqimida qatnashadigan element tushunchasi.
Boolean attribute	Mavjudligining o'zi true ma'nosini beradigan atribut, masalan required.
Browser	HTML/CSS/JS ni talqin qilib web-sahifani ko'rsatadigan dastur.
Canonical	Bir xil/o'xshash URLlar orasidagi afzal manzilni bildiruvchi link rel.
Caption	Jadval yoki figure mazmunini izohlovchi ko'rinadigan sarlavha/izoh.
Checkbox	Bir nechta mustaqil variantni tanlashga mos form control.
Client-side validation	Brauzer tomonida forma qiymatini tekshirish; server tekshiruvini almashtirmaydi.
Closing tag	Elementni yakunlovchi </...> tegi.
Content	Element ichidagi matn, media yoki ichki elementlar.
data-*	Loyiha uchun maxsus metadata saqlashga mo'ljallangan global atributlar.
Datalist	Inputga tavsiya variantlarini beruvchi element.
Debugging	Xatoni tizimli topish, sababini aniqlash va tuzatish jarayoni.
DOCTYPE	Brauzerga hujjatning HTML standard mode da talqin qilinishini bildiruvchi deklaratsiya.

ATAMA	IZOH
DOM	Brauzer HTML hujjatini tugunlar daraxti sifatida ifodalovchi model.
Element	Teg(lar), atributlar va kontentdan tashkil topadigan HTML birligi.
Entity	HTML sintaksisida maxsus bo'lgan belgilarni matn sifatida yozish kodi, masalan <.
Fallback	Asosiy resurs ishlamasa ko'rsatiladigan zaxira variant.
Fieldset	Formada bog'liq boshqaruvlarni semantik guruhlovchi element.
Form control	Input, select, textarea, button kabi foydalanuvchi bilan ishlaydigan forma elementi.
GET	Forma qiymatlari ko'pincha URL query qismida yuboriladigan HTTP usuli.
Heading hierarchy	h1-h6 orqali mazmun bo'limlari o'rtasidagi mantiqiy ierarxiya.
Hidden input	Foydalanuvchiga ko'rinmay, forma bilan yuborilishi mumkin bo'lgan qiymat maydoni.
HTML	Web-sahifa strukturasi va ma'nosini belgilash tili.
HTTP	Brauzer va server o'rtasida web resurslarni uzatish protokoli.
id	Hujjat ichida elementni noyob aniqlash uchun global atribut.
iframe	Boshqa hujjatni joriy sahifa ichida alohida browsing contextda ko'rsatadigan element.
Inline content	Matn oqimi ichida qatnashadigan elementlar tushunchasi.
Label	Forma controlning ko'rinadigan va accessible nomini bog'lashga yordam beruvchi element.
Landmark	main, nav, header kabi sahifaning yirik ma'noli hududi.
Lazy loading	Pastdagi rasm/iframe ni ehtiyojga yaqin vaqtda yuklash strategiyasi.
Legend	Fieldset guruhining ko'rinadigan nomi.
Metadata	Sahifa haqida brauzer, qidiruv tizimi yoki ijtimoiy platformaga beriladigan ma'lumot.
Nesting	Bir HTML elementini boshqasining ichiga joylashtirish.
Open Graph	Link ulashilganda social preview metadata sini ifodalovchi property lar to'plami.
Opening tag	Element boshlanishini bildiruvchi <...> teg.
Pattern	Text input qiymatini regex ko'rinishidagi andoza bilan tekshiruvchi atribut.
POST	Forma ma'lumotini request body orqali yuborishda ko'p ishlatiladigan HTTP usuli.

ATAMA	IZOH
Relative path	Joriy fayl joylashuviga nisbatan yozilgan resurs manzili.
Responsive image	Qurilma/format sharoitiga qarab mos manba tanlanadigan rasm yondashuvi.
Robots meta	Qidiruv crawlerlari uchun indekslash/follow bo'yicha ko'rsatma.
Sandbox	Iframe imkoniyatlarini cheklash uchun xavfsizlik atributi.
Semantic HTML	Element nomi orqali kontent vazifasini ifodalovchi markup.
SEO	Qidiruv tizimlari sahifani tushunishi va topishi uchun texnik/kontent amaliyotlari.
Server-side validation	Server qabul qilgan ma'lumotni ishonchli qayta tekshirish jarayoni.
Source element	Audio/video/picture uchun alternativ resurs manbalarini belgilaydi.
srcset	Bir yoki bir nechta rasm nomzodlarini brauzerga taklif qiluvchi atribut.
Tag	HTML elementini belgilash uchun <...> ko'rinishidagi sintaktik belgi.
Track	Video/audio bilan bog'liq subtitle/caption VTT resursini ulash elementi.
URL	Internet yoki loyiha ichidagi resurs manzili.
UTF-8	Ko'plab tillar va belgilarni ifodalovchi keng tarqalgan character encoding.
Validation	Markup yoki input belgilangan qoidalarga mosligini tekshirish.
Validator	HTML standart xatolarini avtomatik tekshiruvchi vosita.
Viewport	Brauzerda sahifaning ko'rinadigan ekran hududi.
Void element	Yopuvchi tegga ega bo'lmagan HTML element, masalan img yoki br.
WebVTT	Video subtitrlari/captionlari uchun .vtt matn formati.
Wireframe	Sahifa tarkibi va hududlarini dizayndan oldin ko'rsatadigan soddalashtirilgan reja.

HTML teglar indeksi

Bu indeks tegni qaysi chapterda asosiy mavzu sifatida topishingizni tez ko'rsatadi. Ayrim teglar keyingi chapterlarda yana ishlatiladi.

TEG	CHAPTER	ASOSIY VAZIFA
<a>	05, 11, 15	havola
<address>	03	HTML strukturasi/semantikasi
<article>	11, 16	mustaqil kontent
<aside>	11, 16	HTML strukturasi/semantikasi
<audio>	12	audio media
<blockquote>	03	HTML strukturasi/semantikasi
<body>	02	HTML strukturasi/semantikasi
 	01, 03	HTML strukturasi/semantikasi
<button>	09, 10, 15	HTML strukturasi/semantikasi
<caption>	08, 15	HTML strukturasi/semantikasi
<code>	04	HTML strukturasi/semantikasi
<datalist>	10	HTML strukturasi/semantikasi
<dd>	07	HTML strukturasi/semantikasi
	04	HTML strukturasi/semantikasi
<details>	11	HTML strukturasi/semantikasi
<div>	10, 11	HTML strukturasi/semantikasi
<dl>	07	HTML strukturasi/semantikasi
<dt>	07	HTML strukturasi/semantikasi
	04	HTML strukturasi/semantikasi
<embed>	13	HTML strukturasi/semantikasi
<fieldset>	09-10, 15	HTML strukturasi/semantikasi
<figcaption>	06, 16	HTML strukturasi/semantikasi
<figure>	06, 16	HTML strukturasi/semantikasi
<footer>	11, 16	HTML strukturasi/semantikasi
<form>	09-10, 15-16	forma konteyneri
<h1-h6>	03, 15-16	HTML strukturasi/semantikasi
<head>	02, 14	HTML strukturasi/semantikasi

TEG	CHAPTER	ASOSIY VAZIFA
<header>	11, 16	HTML strukturasi/semantikasi
<hr>	01	HTML strukturasi/semantikasi
<html>	02	HTML strukturasi/semantikasi
<iframe>	13	ichki/tashqi hujjat
	06, 15	rasm
<input>	09-10, 15	forma maydoni
<ins>	04	HTML strukturasi/semantikasi
<kbd>	04	HTML strukturasi/semantikasi
<label>	09-10, 15	HTML strukturasi/semantikasi
<legend>	09-10, 15	HTML strukturasi/semantikasi
	07	HTML strukturasi/semantikasi
<link>	14	HTML strukturasi/semantikasi
<main>	11, 15-16	asosiy mazmun
<mark>	04	HTML strukturasi/semantikasi
<meta>	02, 14	metadata
<nav>	05, 11, 16	navigatsiya
<object>	13	HTML strukturasi/semantikasi
	07	HTML strukturasi/semantikasi
<option>	09	HTML strukturasi/semantikasi
<p>	01, 03	HTML strukturasi/semantikasi
<picture>	06	HTML strukturasi/semantikasi
<pre>	03-04	HTML strukturasi/semantikasi
<q>	03	HTML strukturasi/semantikasi
<section>	11, 16	tematik bo'lim
<select>	09-10	HTML strukturasi/semantikasi
<small>	04	HTML strukturasi/semantikasi
<source>	06, 12	HTML strukturasi/semantikasi
	10	HTML strukturasi/semantikasi
	04	HTML strukturasi/semantikasi

TEG	CHAPTER	ASOSIY VAZIFA
<sub>	04	HTML strukturasi/semantikasi
<summary>	11	HTML strukturasi/semantikasi
<sup>	04	HTML strukturasi/semantikasi
<table>	08, 15-16	jadval
<tbody>	08	HTML strukturasi/semantikasi
<td>	08	HTML strukturasi/semantikasi
<textarea>	09-10	HTML strukturasi/semantikasi
<tfoot>	08	HTML strukturasi/semantikasi
<th>	08, 15	HTML strukturasi/semantikasi
<thead>	08	HTML strukturasi/semantikasi
<time>	11, 16	HTML strukturasi/semantikasi
<title>	02, 14	HTML strukturasi/semantikasi
<tr>	08	HTML strukturasi/semantikasi
<track>	12	HTML strukturasi/semantikasi
	07	HTML strukturasi/semantikasi
<video>	12	video media

Debugging va validatsiya checklist

1. FAYL

- Fayl haqiqatan .html mi? O'zgarish saqlanganmi? To'g'ri faylni brauzerda ochdingizmi?

2. SKELET

- DOCTYPE, html lang, head, charset, viewport, title va body joyidami?

3. NESTING

- Ichkarida ochilgan elementlar to'g'ri tartibda yopilganmi? Brauzer DOMni kutilmaganda o'zgartirmaganmi?

4. PATH

- Rasm/link/media yo'li joriy faylga nisbatan to'g'rimi? Fayl nomi katta-kichik harf bilan mosmi?

5. IDS

- id qiymatlari noyobmi? #anchor maqsad id bilan aynan mosmi?

6. FORMS

- Har input name va labelga egami? Radio guruhlar name ni bo'lishadimi? Kerakli maydonlar requiredmi?

7. ACCESSIBILITY

- Alt, heading ierarxiyasi, table scope, native button/link va skip-link tekshirildimi?

8. METADATA

- Har sahifa title/descriptionga egami? Canonical/robots ataylab qo'yilganmi?

9. VALIDATOR

- HTML validator ko'rsatgan errorlarni sababini tushunib tuzatdingizmi? Warning va errorni farqladingizmi?

10. DEVTOOLS

- Elements panelida real DOM, link targetlari va resurs pathlari tekshirildimi?

Starter assetlar

Kitobdagi rasm, audio, video, subtitr va PDF bilan bog'liq challenge larni darhol sinash uchun ushbu kitob bilan birga "HTML_BY_EXAMPLE_STARTER_ASSETS.zip" paketi beriladi. Undagi fayllar o'quv placeholderlari bo'lib, real loyihada o'z kontentingiz bilan almashtiriladi.

[Tavsiya etilgan papka tuzilmasi](#)

```
project/  
  index.html  
  images/  
    hero.jpg  
    hero.webp  
    avatar.jpg  
    poster.jpg  
  media/  
    lesson.mp3  
    lesson.ogg  
    lesson.mp4  
    lesson.webm  
    lesson-uz.vtt  
  files/  
    guide.pdf  
    transcript.txt
```

STARTER ASSETLARNI 4 QADAMDA ISHGA TUSHIRING

- | | |
|--|---|
| 1. ZIP faylni oching | 3. src/href yo'llarini tekshiring |
| 2. Papkalarni project/ ichiga ko'chiring | 4. index.html ni brauzerda ochib sinang |

30 kunlik o‘qish rejasi

Maqsad: kitobni “tez tugatish” emas, har bosqichda kod yozish, tekshirish va capstone darajasiga tayyor bo‘lish. Kunlarni o‘z jadvalingizga mos surishingiz mumkin.

KUN	QAMROV	FOKUS	NATIJA
1-4	Chapter 01-02	HTML asoslari, hujjat skeleti, debugging	Har kuni 20-30 min kod + 1 Quick Check
5-8	Chapter 03-05	Matn, semantika, link va path	3 sahifali mini sayt navigatsiyasi
9-12	Chapter 06-08	Rasm, ro'yxat, jadval	Galereya + accessible jadval
13-17	Chapter 09-10	Oddiy va murakkab formalar	To'liq onboarding formasi
18-21	Chapter 11-12	Semantic arxitektura va media	Blog skeleti + media dars
22-24	Chapter 13-14	Embed xavfsizligi, SEO va social metadata	Resurslar sahifasi + 3 xil head
25-27	Chapter 15-16	Accessibility, refaktor, validator, DevTools	A11y audit + integration workshop
28	Challenge 146-147	Capstone rejalash va 1-loyiha	Validator + checklist bilan topshirish
29	Challenge 148-149	E-commerce yoki blog capstone	Barcha sahifalarda QA
30	Challenge 150	FINAL - Nova Studio	5+ sahifa, yakuniy audit va self-review

Capstone baholash rubrikasi - 100 ball

146-150 loyihalarni o'zingiz yoki ustoz bilan baholash uchun. 80+ ball - mustahkam; 90+ - professional boshlang'ich daraja; 95+ - juda kuchli topshirish.

MEZON	BALL	NIMA TEKSHIRILADI
HTML5 strukturalari va validator	20	DOCTYPE/lang/charset/viewport, toza nesting, katta markup xatosi yo'q.
Semantic arxitektura	15	Landmarklar, heading ierarxiyasi, section/article tanlovi mantiqiy.
Accessibility	20	Alt, label, fieldset/legend, scope, native controls, keyboard/skip link.
Navigatsiya va file path	10	Broken link yo'q, barcha sahifalarda izchil nav, relative pathlar to'g'ri.
Form/jadval/media sifati	10	Talab qilingan komponentlar semantik va foydalanishga qulay.
SEO va metadata	10	Har sahifada noyob title/description; kerak bo'lsa canonical/OG.
Kod o'qilishi	10	Indentation, ma'noli id/class, ortiqcha markup yo'q, kommentlar me'yorida.
Yakuniy QA	5	DevTools, validator va checklist bilan yakuniy tekshiruv bajarilgan.

Rasmiy manbalar va keyingi o'qish

Kitob amaliy o'rganish uchun yozilgan. Standart tafsiloti, element/atribut ma'lumotlari va tekshiruvlar uchun quyidagi manbalardan foydalaning. Havolalar 2026-yil sentabrda tekshirildi.

WHATWG - HTML Living Standard

HTMLning amaldagi Living Standard spetsifikatsiyasi.

[Rasmiy sahifani ochish](https://html.spec.whatwg.org/)

<https://html.spec.whatwg.org/>

WHATWG - Web Developers Edition

Spetsifikatsiyaning web dasturchilar uchun o'qishga qulay nashri.

[Rasmiy sahifani ochish](https://html.spec.whatwg.org/dev/)

<https://html.spec.whatwg.org/dev/>

MDN - HTML Reference

HTML elementlari, atributlari va misollar uchun amaliy reference.

[Rasmiy sahifani ochish](https://developer.mozilla.org/en-US/docs/Web/HTML/Reference)

<https://developer.mozilla.org/en-US/docs/Web/HTML/Reference>

W3C - HTML Checker (Nu)

Zamonaviy HTML hujjatlarini conformance/markup xatolari bo'yicha tekshirish.

[Rasmiy sahifani ochish](https://validator.w3.org/nu/)

<https://validator.w3.org/nu/>

W3C WAI - Tutorials

Rasm, jadval, forma va boshqa accessibility mavzulari bo'yicha rasmiy qo'llanmalar.

[Rasmiy sahifani ochish](https://www.w3.org/WAI/tutorials/)

<https://www.w3.org/WAI/tutorials/>

Tavsiya: kitobdagi misol va challenge ni bajargach, kerakli tegni MDN/WHATWG'da tekshirish va yakunda W3C HTML Checker orqali validatsiya qilish odatini yarating.

PART III

JAVOBLAR

Quick Check + 001-150 • faqat mustaqil urinishdan keyin solishtiring

Quick Check javob kaliti

Chapter 01 — HTML asoslari

1. Web-sahifa mazmuni va strukturasi uchun.
2. Teg - markup belgisi; element - teg(lar), atribut va kontentdan tashkil topgan birlik.
3. Masalan `<img>`, `<br>`, `<hr>`.

Chapter 02 — HTML hujjati, head/body va debugging

1. Brauzer, qidiruv tizimi va metadata iste'molchilari uchun.
2. Sahifaning asosiy tilini bildiradi.
3. Fayl saqlanganmi, teg/nesting to'g'rimi, atribut/path to'g'rimi.

Chapter 03 — Sarlavhalar, paragraf va matn bloklari

1. h1 bosh mavzu, h2 uning bo'limi.
2. q - qisqa inline, blockquote - alohida katta iqtibos.
3. Bo'sh joy va satrlarni asosan saqlaydi.

Chapter 04 — Semantik matn, kod va maxsus belgilar

1. strong muhimlik semantikasini bildiradi, b esa stilistik ajratishdir.
2. O'chirilgan va qo'shilgan/tahrirlangan mazmunda.
3. < belgisi markup sifatida talqin qilinmasligi uchun.

Chapter 05 — Havolalar, fayl yo'llari va navigatsiya

1. Bir papka yuqoriga chiqishni.
2. href="#id" id atributiga ulanadi.
3. Yo'q, nav muhim navigatsiya bloklari uchun.

Chapter 06 — Rasmlar, figure, picture va responsive manbalar

1. alt="".
2. Yo'q: alt rasm o'rnini bosadi, caption qo'shimcha izoh beradi.
3. Fallback va real img element sifatida.

Chapter 07 — Ro'yxatlar va menyu tuzilmalari

1. Tartib/qadam muhim bo'lganda.
2. Tegishli li ichida.
3. dt termin, dd ta'rif/qiyamat.

Chapter 08 — Jadvallar: struktura, birlashtirish va accessibility

1. Jadval maqsadini ko'rinadigan tarzda tushuntiradi.
2. Qator sarlavhasi bo'lgan th da.
3. Katak ikki ustun bo'ylab cho'ziladi.

Chapter 09 — Formalar: input, label va foydalanuvchi ma'lumoti

1. id DOM/label bog'lanishi, name yuboriladigan kalit.
2. Bitta guruhdan faqat bitta tanlov bo'lishi uchun.
3. Yo'q, server yana tekshiradi.

Chapter 10 — Murakkab formalar, fieldset va umumiy atributlar

1. Bog'liq form controllarni guruhlash va guruh nomini berish uchun.
2. Datalist tavsiya beradi, select variantlar ro'yxatidan tanlatadi.
3. Yo'q, DOMda ko'rinishi mumkin.

Chapter 11 — Semantic HTML va sahifa arxitekturas

1. Article mustaqil kontent, section tematik guruh.
2. Ha, bog'liq qo'shimcha kontent sifatida.
3. Kod o'qilishi, SEO/strukturani tushunish, accessibility.

Chapter 12 — Audio, video, source va subtitrlar

1. Brauzer qo'llaydigan birinchi mos manbani tanlaydi.
2. Subtitle tilini bildiradi.
3. Subtitle media vaqtiga bog'liq; transcript to'liq matn ko'rinishidir.

Chapter 13 — iframe, object, embed va xavfsiz tashqi kontent

1. Ekran o'quvchi foydalanuvchilari uchun.
2. Ko'p imkoniyatlarni maksimal cheklaydi.
3. Ichida fallback HTML berish mumkin.

Chapter 14 — head, metadata, SEO va social preview

1. Takroriy/o'xshash URLlarda afzal URLni ko'rsatishda.
2. Sahifani indekslamalik ko'rsatmasi.
3. Yo'q.

Chapter 15 — Accessibility: semantic, alt, label va klaviatura

1. alt="".
2. Native klaviatura/fokus/semantika tayyor.
3. Takroriy navigatsiyani tez chetlab o'tish.

Chapter 16 — Integration, refaktor va debugging workshop

1. Brauzer parseri xato markupni tuzatib DOMni o'zgartirishi mumkin.
2. Hujjat skeleti va semantik ierarxiyadan.
3. Masalan title, nav, heading, alt, label, validator, viewport.

Challenge javoblari

Quyidagi kodlar mumkin bo'lgan yechimlardan biridir. Ishlaydigan, semantik va accessibility jihatdan to'g'ri boshqa yechim ham qabul qilinadi.

Answers — 001-010

001 JAVOB

```
<h1>Mening birinchi saytim</h1>
```

002 JAVOB

```
<p>Jasur</p>
```

003 JAVOB

```
<h1>Men haqimda</h1>
<p>Men HTML o'rganayapman.</p>
<p>Bu mening mashq sahifam.</p>
```

004 JAVOB

```
<p>Frontend<br>HTML asoslari</p>
```

005 JAVOB

```
<p>Birinchi bo'lim</p>
<hr>
<p>Ikkinchi bo'lim</p>
```

006 JAVOB

```
<!-- Bu sarlavha keyin o'zgartiriladi -->
```

007 JAVOB

```
<h1>HTML darsi</h1>
<h2>Bugungi mavzu</h2>
<p>Asosiy HTML teglarini o'rganamiz.</p>
```

008 JAVOB

```
<p>5 &lt; 10 va 10 &gt; 5</p>
```

009 JAVOB

```
<p>&copy; 2026 HTML Book</p>
```

010 JAVOB

```
<h1>Men haqimda</h1>
<p>Men web dasturlashni o'rganayapman.</p>
<hr>
<h2>Maqsadim</h2>
<p>Chiroyli va to'g'ri web-sahifalar yaratish.</p>
```

Answers — 011-020**011 JAVOB**

```
<!DOCTYPE html>
<html>
<head>
  <title>Sahifa</title>
</head>
<body>
</body>
</html>
```

012 JAVOB

```
<html lang="uz">
</html>
```

013 JAVOB

```
<meta charset="UTF-8">
```

014 JAVOB

```
<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <title>HTML darsligi</title>
</head>
<body></body>
</html>
```

015 JAVOB

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

016 JAVOB

```
<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <title>Mashq</title>
</head>
<body>
  <h1>Mashq sahifasi</h1>
</body>
</html>
```

017 JAVOB

```
<body>
  <h1>Sarlavha</h1>
  <p>Matn.</p>
</body>
```

018 JAVOB

```
<head>
  <!-- Muallif: Talaba -->
  <title>Sahifa</title>
</head>
```

019 JAVOB

```
<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <title>To'g'ri hujjat</title>
</head>
<body>
  <p>Tayyor.</p>
</body>
</html>
```

020 JAVOB

```
<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Mening profilim</title>
</head>
<body>
  <h1>Mening profilim</h1>
  <p>Bu mening shaxsiy sahifam.</p>
</body>
```

```
</html>
```

Answers — 021-030

021 JAVOB

```
<h1>Dasturlash</h1>
<h2>Frontend</h2>
<h3>HTML</h3>
```

022 JAVOB

```
<p>HTML sahifa tuzilishini belgilaydi.</p>
<p>CSS sahifaning ko'rinishini bezaydi.</p>
```

023 JAVOB

```
<h1>1-daraja</h1>
<h2>2-daraja</h2>
<h3>3-daraja</h3>
<h4>4-daraja</h4>
<h5>5-daraja</h5>
<h6>6-daraja</h6>
```

024 JAVOB

```
<p>Ustoz aytdi: <q>Har kuni kod yozing.</q></p>
```

025 JAVOB

```
<blockquote>
  Katta natija kichik, ammo muntazam qadamlar orqali keladi.
</blockquote>
```

026 JAVOB

```
<pre>HTML
  CSS
  JavaScript</pre>
```

027 JAVOB

```
<address>
  TATU talabasi<br>
  Toshkent, O'zbekiston
</address>
```

028 JAVOB

```
<h1>Internet tarixi</h1>
<h2>Web paydo bo'lishi</h2>
<p>Web axborotni bog'lash uchun yaratildi.</p>
<p>HTML esa web-sahifalarni belgilash uchun ishlatiladi.</p>
```

029 JAVOB

```
<h2>Nazariya</h2>
<p>Avval tushunamiz.</p>
<hr>
<h2>Amaliyot</h2>
<p>Keyin kod yozamiz.</p>
```

030 JAVOB

```
<h1>Web texnologiyalar</h1>
<h2>Frontend</h2>
<p>Frontend foydalanuvchi ko'radigan qismdir.</p>
<h3>HTML</h3>
<p>HTML strukturani yaratadi.</p>
<h3>CSS</h3>
<p>CSS dizaynni boshqaradi.</p>
```

Answers — 031-040

031 JAVOB

```
<p><strong>Diqqat:</strong> faylni saqlang.</p>
```

032 JAVOB

```
<p>HTMLni <em>amaliyot</em> orqali tez o'rganasiz.</p>
```

033 JAVOB

```
<p>Bugungi mavzu: <mark>HTML teglar</mark>.</p>
```

034 JAVOB

```
<p><del>120 000 so'm</del> <ins>99 000 so'm</ins></p>
```

035 JAVOB

```
<p>H<sub>2</sub></sub>0</p>
```

036 JAVOB

```
<p>x<sup>2</sup></p>
```

037 JAVOB

```
<p>Narx: 50 000 so'm <small>Soliqlar bilan.</small></p>
```

038 JAVOB

```
<p>Saqlash uchun <kbd>Ctrl</kbd> + <kbd>S</kbd> bosing.</p>
```

039 JAVOB

```
<p>Paragraf uchun <code>&lt;p&gt;</code> tegidan foydalaniladi.</p>
```

040 JAVOB

```
<p><strong>Eslatma:</strong> <em>bugun</em> <mark>paragraf</mark> uchun <code>&lt;p&gt;</code>
tegini mashq qiling.</p>
```

Answers — 041-050

041 JAVOB

```
<a href="about.html">Biz haqimizda</a>
```

042 JAVOB

```
<a href="https://example.com" target="_blank" rel="noopener">Example</a>
```

043 JAVOB

```
<a href="contact.html">Bog'lanish</a>
```

044 JAVOB

```
<a href="mailto:hello@example.com">Email yuborish</a>
```

045 JAVOB

```
<a href="tel:+998901234567">+998 90 123 45 67</a>
```

046 JAVOB

```
<a href="#services">Xizmatlarga o'tish</a>
<h2 id="services">Xizmatlar</h2>
```

047 JAVOB

```
<a href="files/guide.pdf">Qo'llanmani ochish</a>
```

048 JAVOB

```
<a href="files/guide.pdf" download>PDFni yuklab olish</a>
```

049 JAVOB

```
<nav>
  <a href="index.html">Bosh sahifa</a>
  <a href="courses.html">Kurslar</a>
  <a href="contact.html">Kontakt</a>
</nav>
```

050 JAVOB

```
<body id="top">
  <a href="#bottom">Pastga</a>
  <p>Kontent...</p>
  <footer id="bottom">
    <a href="#top">Yuqoriga</a>
  </footer>
</body>
```

Answers — 051-060**051 JAVOB**

```

```

052 JAVOB

```

```

053 JAVOB

```

```

054 JAVOB

```
<figure>
  
  <figcaption>HTML kodining oddiy namunasi.</figcaption>
</figure>
```

055 JAVOB

```
<h1>Mening profilim</h1>

```

056 JAVOB

```
<a href="home.html">
  
</a>
```

057 JAVOB

```
<picture>
  <source srcset="images/hero.webp" type="image/webp">
  
</picture>
```

058 JAVOB

```
<picture>
  <source media="(max-width: 600px)" srcset="images/mobile.jpg">
  <source media="(min-width: 601px)" srcset="images/desktop.jpg">
  
</picture>
```

059 JAVOB

```

```

060 JAVOB

```
<figure>
  
  <figcaption>Birinchi manzara</figcaption>
</figure>
<figure>
  
  <figcaption>Ikkinchi manzara</figcaption>
</figure>
<figure>
  
  <figcaption>Uchinchi manzara</figcaption>
</figure>
```

Answers — 061-070

061 JAVOB

```
<ul>
  <li>Frontend</li>
  <li>Backend</li>
  <li>Mobile</li>
</ul>
```

062 JAVOB

```
<ol>
  <li>Fayl yarating</li>
  <li>Kod yozing</li>
  <li>Saqlang</li>
  <li>Brauzerda oching</li>
</ol>
```

063 JAVOB

```
<ol start="5">
  <li>Beshinchi</li>
  <li>Oltinchi</li>
</ol>
```

064 JAVOB

```
<ol reversed>
  <li>Final</li>
  <li>Yarim final</li>
  <li>Chorak final</li>
</ol>
```

065 JAVOB

```
<ul>
  <li>Frontend
    <ul>
      <li>HTML</li>
      <li>CSS</li>
    </ul>
  </li>
</ul>
```

066 JAVOB

```
<dl>
  <dt>HTML</dt>
  <dd>Sahifa tuzilishini belgilaydi.</dd>
  <dt>CSS</dt>
  <dd>Sahifa ko'rinishini boshqaradi.</dd>
</dl>
```

067 JAVOB

```
<nav>
  <ul>
    <li><a href="index.html">Bosh sahifa</a></li>
    <li><a href="about.html">Biz haqimizda</a></li>
    <li><a href="contact.html">Kontakt</a></li>
  </ul>
</nav>
```

068 JAVOB

```
<h2>Xaridlar</h2>
<ul>
  <li>Non</li>
  <li>Sut</li>
  <li>Meva</li>
</ul>
```

069 JAVOB

```
<ol>
  <li>HTML
    <ul><li>Teglar</li><li>Formalar</li></ul>
  </li>
</ol>
```

```

</li>
<li>CSS
  <ul><li>Selektorlar</li><li>Layout</li></ul>
</li>
<li>JavaScript</li>
</ol>

```

070 JAVOB

```

<h2>Ingredientlar</h2>
<ul>
  <li>2 ta tuxum</li>
  <li>Un</li>
  <li>Sut</li>
</ul>
<h2>Tayyorlash</h2>
<ol>
  <li>Masalliqnlarni aralashtiring.</li>
  <li>Tovada pishiring.</li>
</ol>

```

Answers — 071-080

071 JAVOB

```

<table>
  <tr><th>Til</th><th>Daraja</th></tr>
  <tr><td>HTML</td><td>Boshlang'ich</td></tr>
  <tr><td>CSS</td><td>Boshlang'ich</td></tr>
</table>

```

072 JAVOB

```

<table>
  <caption>Haftalik darslar</caption>
  <tr><th>Kun</th><th>Mavzu</th></tr>
  <tr><td>Dushanba</td><td>HTML</td></tr>
</table>

```

073 JAVOB

```

<table>
  <tr>
    <th scope="col">Ism</th>
    <th scope="col">Ball</th>
  </tr>
  <tr><td>Aziz</td><td>91</td></tr>
</table>

```

074 JAVOB

```

<table>
  <thead>
    <tr><th>Mahsulot</th><th>Narx</th></tr>
  </thead>
  <tbody>
    <tr><td>Kitob</td><td>80 000</td></tr>
  </tbody>
</table>

```

075 JAVOB

```

<table>
  <thead><tr><th>Mahsulot</th><th>Narx</th></tr></thead>
  <tbody><tr><td>Kitob</td><td>80 000</td></tr></tbody>
  <tfoot><tr><th>Jami</th><td>80 000</td></tr></tfoot>
</table>

```

076 JAVOB

```
<table>
  <caption>Guruh natijalari</caption>
  <thead><tr><th scope="col">Guruh</th><th scope="col">Talaba</th><th
scope="col">Ball</th></tr></thead>
  <tbody>
    <tr><th colspan="3">A guruh</th></tr>
    <tr><td>A</td><td>Ali</td><td>92</td></tr>
    <tr><th colspan="3">B guruh</th></tr>
    <tr><td>B</td><td>Madina</td><td>89</td></tr>
  </tbody>
</table>
```

077 JAVOB

```
<table>
  <tr><th scope="col">Fan</th><th scope="col">Guruh</th><th scope="col">Vaqt</th></tr>
  <tr><th scope="rowgroup" rowspan="2">HTML</th><td>A</td><td>09:00</td></tr>
  <tr><td>B</td><td>11:00</td></tr>
</table>
```

078 JAVOB

```
<table>
  <caption>Dars jadvali</caption>
  <thead><tr><th scope="col">Kun</th><th scope="col">Vaqt</th><th scope="col">Fan</th><th
scope="col">Xona</th></tr></thead>
  <tbody>
    <tr><td>Dushanba</td><td>09:00</td><td>HTML</td><td>201</td></tr>
    <tr><td>Seshanba</td><td>10:30</td><td>CSS</td><td>202</td></tr>
    <tr><td>Chorshanba</td><td>09:00</td><td>Git</td><td>203</td></tr>
    <tr><td>Juma</td><td>14:00</td><td>Amaliyot</td><td>Lab</td></tr>
  </tbody>
</table>
```

079 JAVOB

```
<table>
  <caption>Talaba natijalari</caption>
  <thead><tr><th scope="col">Talaba</th><th scope="col">HTML</th><th
scope="col">CSS</th></tr></thead>
  <tbody>
    <tr><th scope="row">Ali</th><td>90</td><td>86</td></tr>
    <tr><th scope="row">Malika</th><td>94</td><td>91</td></tr>
  </tbody>
</table>
```

080 JAVOB

```
<table>
  <caption>Kurs materiallari narxi</caption>
  <thead><tr><th scope="col">Mahsulot</th><th scope="col">Soni</th><th
scope="col">Narx</th></tr></thead>
  <tbody>
    <tr><th scope="row">HTML kitobi</th><td>1</td><td>80 000</td></tr>
    <tr><th scope="row">Daftar</th><td>2</td><td>20 000</td></tr>
    <tr><th scope="row">Marker</th><td>1</td><td>12 000</td></tr>
    <tr><th scope="row">Flash drive</th><td>1</td><td>70 000</td></tr>
  </tbody>
  <tfoot><tr><th colspan="2" scope="row">Jami</th><td>182 000</td></tr></tfoot>
</table>
```

Answers — 081-090

081 JAVOB

```
<label for="name">Ism</label>
```

```
<input id="name" name="name" type="text">
```

082 JAVOB

```
<label for="email">Email</label>
<input id="email" name="email" type="email" required>
```

083 JAVOB

```
<label for="password">Parol</label>
<input id="password" name="password" type="password">
```

084 JAVOB

```
<label for="age">Yosh</label>
<input id="age" name="age" type="number" min="1" max="120">
```

085 JAVOB

```
<fieldset>
  <legend>Rol</legend>
  <label><input type="radio" name="role" value="student"> Talaba</label>
  <label><input type="radio" name="role" value="teacher"> O'qituvchi</label>
</fieldset>
```

086 JAVOB

```
<fieldset>
  <legend>Qiziqishlar</legend>
  <label><input type="checkbox" name="interest" value="html"> HTML</label>
  <label><input type="checkbox" name="interest" value="css"> CSS</label>
</fieldset>
```

087 JAVOB

```
<label for="city">Shahar</label>
<select id="city" name="city">
  <option value="tashkent">Toshkent</option>
  <option value="bukhara">Buxoro</option>
  <option value="navoi">Navoiy</option>
</select>
```

088 JAVOB

```
<label for="message">Xabar</label>
<textarea id="message" name="message" rows="5" cols="40"></textarea>
```

089 JAVOB

```
<form action="/contact" method="post">
  <label for="name">Ism</label>
  <input id="name" name="name" type="text" required>
  <label for="email">Email</label>
  <input id="email" name="email" type="email" required>
  <label for="subject">Mavzu</label>
  <select id="subject" name="subject" required><option
value="">Tanlang</option><option>Taklif</option><option>Savol</option></select>
  <label for="message">Xabar</label>
  <textarea id="message" name="message" rows="6" required></textarea>
  <button type="submit">Yuborish</button>
</form>
```

090 JAVOB

```

<form action="/register" method="post">
  <label for="first">Ism</label><input id="first" name="first" required>
  <label for="last">Familiya</label><input id="last" name="last" required>
  <label for="email">Email</label><input id="email" name="email" type="email" required>
  <label for="birth">Tug'ilgan sana</label><input id="birth" name="birth" type="date">
  <label for="pass">Parol</label><input id="pass" name="password" type="password" minlength="8"
  required>
  <fieldset><legend>Rol</legend><label><input type="radio" name="role" value="student"
  required> Talaba</label><label><input type="radio" name="role" value="teacher">
  O'qituvchi</label></fieldset>
  <fieldset><legend>Qiziqishlar</legend><label><input type="checkbox" name="interest"
  value="html"> HTML</label><label><input type="checkbox" name="interest" value="css">
  CSS</label></fieldset>
  <label for="city">Shahar</label><select id="city"
  name="city"><option>Toshkent</option><option>Buxoro</option></select>
  <button type="submit">Ro'yxatdan o'tish</button>
</form>

```

Answers — 091-100

091 JAVOB

```

<form>
  <fieldset>
    <legend>Shaxsiy ma'lumotlar</legend>
    <label for="first-name">Ism</label>
    <input id="first-name" name="first_name" type="text" required>
    <label for="last-name">Familiya</label>
    <input id="last-name" name="last_name" type="text" required>
  </fieldset>
</form>

```

092 JAVOB

```

<label for="email">Email</label>
<input id="email" name="email" type="email" autocomplete="email" placeholder="siz@example.com">
<label for="phone">Telefon</label>
<input id="phone" name="phone" type="tel" autocomplete="tel" placeholder="+998 90 123 45 67">

```

093 JAVOB

```

<fieldset>
  <legend>Rolingiz</legend>
  <label><input type="radio" name="role" value="student" required> Talaba</label>
  <label><input type="radio" name="role" value="teacher"> O'qituvchi</label>
  <label><input type="radio" name="role" value="developer"> Dasturchi</label>
</fieldset>

```

094 JAVOB

```

<fieldset>
  <legend>Qiziqishlar</legend>
  <label><input type="checkbox" name="interest" value="html"> HTML</label>
  <label><input type="checkbox" name="interest" value="css"> CSS</label>
  <label><input type="checkbox" name="interest" value="js"> JavaScript</label>
</fieldset>

```

095 JAVOB

```

<label for="city">Shahar</label>
<input id="city" name="city" list="cities">
<datalist id="cities">
  <option value="Toshkent">
  <option value="Buxoro">
  <option value="Navoiy">
</datalist>

```

096 JAVOB

```
<label for="avatar">Profil rasmi</label>
<input id="avatar" name="avatar" type="file" accept="image/png,image/jpeg">
```

097 JAVOB

```
<label for="username">Username</label>
<input id="username" name="username" pattern="[A-Za-z0-9_]{4,16}" title="4-16 ta lotin harfi,
raqam yoki underscore" required>
<p id="username-help">Masalan: jasur_27</p>
```

098 JAVOB

```
<form action="/apply" method="post">
  <input type="hidden" name="form_version" value="2">
  <section class="form-step" data-step="1"><h2>1. Profil</h2><label
for="name">Ism</label><input id="name" name="name"></section>
  <section class="form-step" data-step="2"><h2>2. Aloqa</h2><label
for="email">Email</label><input id="email" name="email" type="email"></section>
  <button type="submit">Yuborish</button>
</form>
```

099 JAVOB

```
<form action="/survey" method="post">
  <fieldset><legend>Daraja</legend><label><input type="radio" name="level" value="beginner"
required> Boshlovchi</label><label><input type="radio" name="level" value="intermediate">
O'rta</label></fieldset>
  <fieldset><legend>Qiziqishlar</legend><label><input type="checkbox" name="interest"
value="frontend"> Frontend</label><label><input type="checkbox" name="interest"
value="backend"> Backend</label></fieldset>
  <label for="rating">Baholash</label><select id="rating" name="rating" required><option
value="">Tanlang</option><option>5</option><option>4</option></select>
  <label for="comment">Izoh</label><textarea id="comment" name="comment"></textarea>
  <button type="submit">Yuborish</button><button type="reset">Tozalash</button>
</form>
```

100 JAVOB

```
<form action="/register" method="post" enctype="multipart/form-data">
  <input type="hidden" name="form_version" value="2">
  <fieldset><legend>Shaxsiy ma'lumotlar</legend>
    <label for="name">Ism</label><input id="name" name="name" autocomplete="name" required>
    <label for="email">Email</label><input id="email" name="email" type="email"
autocomplete="email" required>
    <label for="username">Username</label><input id="username" name="username" pattern="[A-Za-
z0-9_]{4,16}" required>
    <label for="city">Shahar</label><input id="city" name="city" list="cities"><datalist
id="cities"><option value="Toshkent"><option value="Buxoro"><option value="Navoiy"></datalist>
    <label for="avatar">Profil rasmi</label><input id="avatar" name="avatar" type="file"
accept="image/png,image/jpeg">
  </fieldset>
  <fieldset><legend>Profil</legend>
    <label><input type="radio" name="role" value="student" required>
Talaba</label><label><input type="radio" name="role" value="developer"> Dasturchi</label>
    <label><input type="checkbox" name="interest" value="html"> HTML</label><label><input
type="checkbox" name="interest" value="css"> CSS</label>
  </fieldset>
  <button type="submit">Ro'yxatdan o'tish</button>
</form>
```

Answers — 101-110

101 JAVOB

```
<header>
  <h1>Code Academy</h1>
  <p>Har kuni bir qadam.</p>
</header>
```

102 JAVOB

```
<nav aria-label="Asosiy navigatsiya">
  <a href="index.html">Bosh sahifa</a>
  <a href="courses.html">Kurslar</a>
  <a href="contact.html">Kontakt</a>
</nav>
```

103 JAVOB

```
<main>
  <article>
    <h1>HTML nima?</h1>
    <p>HTML web-sahifa tuzilishini belgilaydi.</p>
  </article>
</main>
```

104 JAVOB

```
<main>
  <section><h2>Kurs haqida</h2><p>HTMLni amaliyot bilan o'rganamiz.</p></section>
  <section><h2>Dastur</h2><p>16 chapter va 150 challenge.</p></section>
  <section><h2>Natijalar</h2><p>Semantic va accessible sahifalar yarata olasiz.</p></section>
</main>
```

105 JAVOB

```
<article>
  <header>
    <h2>HTMLni amaliyot bilan o'rganish</h2>
    <p>Muallif: Ali • <time datetime="2026-09-12">12-sentabr 2026</time></p>
  </header>
  <p>Har kuni kichik sahifa yarating.</p>
  <p>Xatolarni validator va DevTools bilan tekshiring.</p>
  <footer><p>Teglar: HTML, ta'lim</p></footer>
</article>
```

106 JAVOB

```
<aside>
  <h2>0'xshash mavzular</h2>
  <ul>
    <li><a href="forms.html">Formalar</a></li>
    <li><a href="semantic.html">Semantic HTML</a></li>
    <li><a href="ally.html">Accessibility</a></li>
  </ul>
</aside>
```

107 JAVOB

```
<footer>
  <p>&copy; 2026 Code Academy</p>
  <a href="mailto:hello@example.com">hello@example.com</a>
  <a href="tel:+998901234567">+998 90 123 45 67</a>
  <a href="privacy.html">Maxfiylik siyosati</a>
</footer>
```

108 JAVOB

```
<section aria-labelledby="faq-title">
  <h2 id="faq-title">FAQ</h2>
  <details><summary>HTML nima?</summary><p>Web-sahifa tuzilishini belgilash tili.</p></details>
  <details><summary>CSS kerakmi?</summary><p>Dizayn uchun keyingi bosqichda kerak bo'ladi.</p></details>
  <details><summary>JavaScript qachon?</summary><p>Interaktivlik kerak bo'lganda.</p></details>
</section>
```

109 JAVOB

```
<header><h1>HTML kursi</h1></header>
<nav aria-label="Asosiy navigatsiya"><a href="#about">Kurs</a> <a href="#program">Dastur</a> <a href="#register">Yozilish</a></nav>
<main>
  <section id="about"><h2>Kurs haqida</h2><p>Amaliy HTML.</p></section>
  <section id="program"><h2>Dastur</h2><ol><li>Asoslar</li><li>Semantic HTML</li><li>Formalar</li></ol></section>
  <section id="register"><h2>Yozilish</h2><a href="register.html">Ariza topshirish</a></section>
</main>
<footer><p>&copy; 2026 HTML kursi</p></footer>
```

110 JAVOB

```
<!-- Refaktor qilingan semantic variant -->
<header><h1>Tech Blog</h1></header>
<nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a href="archive.html">Arxiv</a></nav>
<main>
  <article>
    <h2>Semantic HTML</h2>
    <p><time datetime="2026-09-12">12-sentabr 2026</time></p>
    <p>Ma'noli elementlar sahifa tuzilishini tushunarli qiladi.</p>
  </article>
  <aside><h2>Tavsiya</h2><a href="ally.html">Accessibilityni o'rganing</a></aside>
</main>
<footer><p>&copy; 2026 Tech Blog</p></footer>
```

Answers — 111-118

111 JAVOB

```
<audio controls src="media/lesson.mp3">
  Brauzeringiz audio elementini qo'llamaydi.
</audio>
```

112 JAVOB

```
<audio controls>
  <source src="media/lesson.mp3" type="audio/mpeg">
  <source src="media/lesson.ogg" type="audio/ogg">
  Audio ochildi.
</audio>
```

113 JAVOB

```
<video controls src="media/intro.mp4" width="720">
  Video ochildi.
</video>
```

114 JAVOB

```
<video controls width="720" poster="images/intro-poster.jpg">
  <source src="media/intro.mp4" type="video/mp4">
</video>
```

115 JAVOB

```
<video controls width="720">
  <source src="media/lesson.webm" type="video/webm">
  <source src="media/lesson.mp4" type="video/mp4">
  Video ochildi.
</video>
```

116 JAVOB

```
<video controls width="720">
  <source src="media/lesson.mp4" type="video/mp4">
  <track src="media/lesson-uz.vtt" kind="subtitles" srclang="uz" label="O'zbekcha" default>
</video>
```

117 JAVOB

```
<audio controls loop>
  <source src="media/music.mp3" type="audio/mpeg">
</audio>
```

118 JAVOB

```
<main>
  <h1>Media dars</h1>
  <section><h2>Audio</h2><audio controls><source src="media/lesson.mp3"
type="audio/mpeg"><source src="media/lesson.ogg" type="audio/ogg">Audio
ochilmadi.</audio></section>
  <section>
    <h2>Video</h2>
    <video controls width="720" poster="images/poster.jpg">
      <source src="media/lesson.webm" type="video/webm">
      <source src="media/lesson.mp4" type="video/mp4">
      <track src="media/lesson-uz.vtt" kind="subtitles" srclang="uz" label="O'zbekcha" default>
    </video>
    <p><a href="files/transcript.txt">Video transcriptini o'qish</a></p>
  </section>
</main>
```

Answers — 119-126**119 JAVOB**

```
<iframe src="about.html" title="Biz haqimizda"></iframe>
```

120 JAVOB

```
<iframe src="map.html" title="Xarita" width="800" height="500"></iframe>
```

121 JAVOB

```
<iframe src="reviews.html" title="Sharhlar" loading="lazy"></iframe>
```

122 JAVOB

```
<iframe src="widget.html" title="Vidget" sandbox></iframe>
```

123 JAVOB

```
<iframe src="form-widget.html" title="Forma vidjeti" sandbox="allow-forms"></iframe>
```

124 JAVOB

```
<embed src="files/book.pdf" type="application/pdf" width="100%" height="700">
```

125 JAVOB

```
<object data="files/book.pdf" type="application/pdf" width="100%" height="700">
  <p>PDF ochilmadi. <a href="files/book.pdf">PDFni yuklab oling</a>.</p>
</object>
```

126 JAVOB

```
<main>
  <h1>Resurslar</h1>
  <section><h2>Ichki demo</h2><iframe src="demo.html" title="HTML demo" loading="lazy"
width="800" height="500"></iframe></section>
  <section><h2>Forma vidjeti</h2><iframe src="form-widget.html" title="Forma vidjeti"
sandbox="allow-forms"></iframe></section>
  <section><h2>PDF qo'llanma</h2><object data="files/guide.pdf" type="application/pdf"
width="100%" height="700"><p>PDF ko'rinmadi. <a href="files/guide.pdf">Qo'llanmani yuklab
oling</a>.</p></object></section>
</main>
```

Answers — 127-134

127 JAVOB

```
<title>HTML kursi</title>
<meta name="description" content="HTML asoslarini amaliy topshiriqlar bilan o'rganing.">
```

128 JAVOB

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

129 JAVOB

```
<link rel="icon" href="/favicon.ico">
```

130 JAVOB

```
<link rel="canonical" href="https://example.com/html-course">
```

131 JAVOB

```
<meta name="robots" content="index, follow">
```

132 JAVOB

```
<meta property="og:title" content="HTML by Example">
<meta property="og:description" content="150 ta challenge bilan HTMLni o'rganing.">
```

133 JAVOB

```
<meta property="og:image" content="https://example.com/images/cover.jpg">
<meta property="og:type" content="website">
```

134 JAVOB

```
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>HTML by Example | Amaliy HTML darsligi</title>
  <meta name="description" content="HTMLni 150 ta bosqichma-bosqich amaliy topshiriq bilan
o'rganing.">
  <meta name="robots" content="index, follow">
  <link rel="canonical" href="https://example.com/html-book">
  <link rel="icon" href="/favicon.ico">
  <meta property="og:title" content="HTML by Example">
  <meta property="og:description" content="150 ta amaliy topshiriqli o'zbekcha HTML darsligi">
  <meta property="og:image" content="https://example.com/images/html-book-cover.jpg">
  <meta property="og:type" content="website">
</head>
```

Answers — 135-140

135 JAVOB

```

```

136 JAVOB

```

```

137 JAVOB

```
<label for="email">Email manzil</label>
<input id="email" name="email" type="email">
```

138 JAVOB

```
<table>
  <tr>
    <th scope="col">Kurs</th>
    <th scope="col">Davomiylik</th>
  </tr>
  <tr><td>HTML</td><td>4 hafta</td></tr>
</table>
```

139 JAVOB

```
<button type="button">Batafsil ko'rish</button>
```

140 JAVOB

```
<a href="#main">Asosiy kontentga o'tish</a>
<header><h1>Kontakt</h1></header>
<main id="main">
  
  
  <form><label for="email">Email</label><input id="email" name="email" type="email"
required><button type="submit">Yuborish</button></form>
  <table><caption>Ish vaqti</caption><tr><th scope="col">Kun</th><th
scope="col">Vaqt</th></tr><tr><th scope="row">Dushanba</th><td>09:00-18:00</td></tr></table>
  <button type="button">Batafsil ko'rish</button>
</main>
```

Answers — 141-145

141 JAVOB

```
<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>To'g'rilangan sahifa</title>
</head>
<body>
  <header><h1>Kurs sahifasi</h1></header>
  <main id="main">
    <section id="about"><h2>Kurs haqida</h2><p>Amaliy HTML kursi.</p></section>
    <section id="apply"><h2>Ariza</h2><form><label for="email">Email</label><input id="email"
name="email" type="email" required><button type="submit">Yuborish</button></form></section>
  </main>
</body>
</html>
```

142 JAVOB

```
<header><h1>Tech Blog</h1></header>
<nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="archive.html">Arxiv</a></nav>
<main>
  <article>
    <header><h2>HTMLni to'g'ri o'rganish</h2><p><time datetime="2026-09-12">12-sentabr
2026</time></p></header>
    <figure><figcaption>Har kuni kichik amaliyot.</figcaption></figure>
    <p>Avval strukturani, keyin semantikani mustahkamlang.</p>
  </article>
  <aside><h2>0'xshash mavzular</h2><a href="ally.html">Accessibility</a></aside>
</main>
<footer><p>&copy; 2026 Tech Blog</p></footer>
```

143 JAVOB

```
<!DOCTYPE html>
<html lang="uz">
<head><meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-
scale=1.0"><title>HTML by Example | Kitob</title><meta name="description" content="Amaliy HTML
darsligi."></head>
<body>
<main><article class="product">
  <h1>HTML by Example</h1>
  <figure><figcaption>150
challenge li darslik</figcaption></figure>
  <p><strong>89 000 so'm</strong></p>
  <h2>Xususiyatlar</h2><ul><li>0'zbek tilida</li><li>150 challenge</li><li>Javoblar
bilan</li></ul>
  <h2>Yetkazish</h2><table><tr><th scope="col">Hudud</th><th
scope="col">Muddat</th></tr><tr><th scope="row">Toshkent</th><td>1 kun</td></tr></table>
  <button type="button">Buyurtma berish</button>
</article></main>
</body></html>
```

144 JAVOB

```
<main>
  <section><h1>HTML kursiga yozilish</h1><p>4 haftalik amaliy
kurs.</p><h2>Tablalar</h2><ul><li>Kompyuter</li><li>Brauzer</li><li>Har kuni
amaliyot</li></ul></section>
  <section><h2>Dars jadvali</h2><table><tr><th scope="col">Kun</th><th
scope="col">Vaqt</th></tr><tr><th
scope="row">Dushanba</th><td>18:00</td></tr></table></section>
  <section><h2>Ariza</h2><form action="/apply" method="post">
    <fieldset><legend>Shaxsiy ma'lumotlar</legend><label for="name">Ism</label><input id="name"
name="name" required><label for="email">Email</label><input id="email" name="email"
type="email" required></fieldset>
    <fieldset><legend>0'qish holati</legend><label><input type="radio" name="level"
value="beginner" required> Boshlovchi</label><label><input type="radio" name="level"
value="basic"> Asoslarni bilaman</label></fieldset>
    <button type="submit">Ariza yuborish</button>
  </form></section>
</main>
```

145 JAVOB

```
<!DOCTYPE html>
<html lang="uz">
<head><meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-
scale=1.0"><title>Jasur | Web portfolio</title><meta name="description" content="Jasurning HTML
loyihalari va aloqa ma'lumotlari."></head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header><h1>Jasur - Portfoliyo</h1><nav aria-label="Asosiy navigatsiya"><a href="#about">Men
haqimda</a> <a href="#skills">Ko'nikmalar</a> <a href="#projects">Loyihalar</a> <a
href="#contact">Kontakt</a></nav></header>
  <main id="main">
    <section id="about"><h2>Men haqimda</h2><figure><figcaption>Web dasturlash talabasi</figcaption></figure><p>Semantic va
accessible web sahifalar yaratishni o'rganaman.</p></section>
    <section id="skills"><h2>Ko'nikmalar</h2><ul><li>HTML5</li><li>Accessibility
asoslari</li><li>SEO metadata</li></ul></section>
```

```

<section id="projects"><h2>Loyihalar</h2><table><caption>Portfolio loyihalari</caption><tr><th
scope="col">Loyiha</th><th scope="col">Holat</th></tr><tr><th
scope="row">Blog</th><td>Tayyor</td></tr><tr><th
scope="row">Shop</th><td>Jarayonda</td></tr></table></section>
<section id="contact"><h2>Kontakt</h2><form><label for="name">Ism</label><input id="name"
name="name" required><label for="email">Email</label><input id="email" name="email"
type="email" required><label for="message">Xabar</label><textarea id="message" name="message"
required></textarea><button type="submit">Yuborish</button></form></section>
</main><footer><p>&copy; 2026 Jasur</p></footer>
</body></html>

```

Yakuniy loyihalar - to'liq yechimlar

Quyidagi variantlarda har bir topshiriqda talab qilingan barcha HTML fayllar ko'rsatilgan. Fayllarni alohida saqlang va navigatsiya linklarini papka ichida sinang.

146 JAVOB — Shaxsiy portfolio

Loyiha fayllari

```
project/
  index.html
  about.html
  projects.html
  contact.html
```

index.html

```
<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Jasur | Portfolio</title>
  <meta name="description" content="Jasurning web portfolio bosh sahifasi.">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Jasur Portfolio</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="about.html">Men haqimda</a> <a href="projects.html">Loyihalar</a> <a
href="contact.html">Kontakt</a></nav>
  </header>
  <main id="main">

    <section><h2>Web dasturlash portfolio</h2><p>Semantic va accessible HTML loyihalarim.</p><a
href="projects.html">Loyihalarni ko'rish</a></section>
    <section><h2>Yo'nalishlar</h2><ul><li>HTML5</li><li>Accessibility</li><li>SEO
asoslari</li></ul></section>
  </main>
  <footer><p>&copy; 2026 Jasur Portfolio</p></footer>
</body>
</html>
```

about.html

```
<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Jasur | Men haqimda</title>
  <meta name="description" content="Jasur haqida va uning web ko'nikmalari.">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Jasur Portfolio</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="about.html">Men haqimda</a> <a href="projects.html">Loyihalar</a> <a
href="contact.html">Kontakt</a></nav>
  </header>
  <main id="main">

    <article><h2>Men haqimda</h2><figure><figcaption>Web dasturlash
talabasi</figcaption></figure><p>HTMLni real loyihalar orqali
o'rganaman.</p><h3>Ko'nikmalar</h3><ul><li>Semantic
```

```

HTML</li><li>Formalar</li><li>Accessibility</li></ul></article>
</main>
<footer><p>&copy; 2026 Jasur Portfolio</p></footer>
</body>
</html>

```

projects.html

```

<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Jasur | Loyihalar</title>
  <meta name="description" content="Jasurning HTML loyihalari ro'yxati.">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Jasur Portfolio</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="about.html">Men haqimda</a> <a href="projects.html">Loyihalar</a> <a
href="contact.html">Kontakt</a></nav>
  </header>
  <main id="main">

    <h2>Loyihalar</h2>
    <article><h3>Universitet sayti</h3><p>Ko'p sahifali semantic website.</p></article>
    <article><h3>Online do'kon</h3><p>Mahsulot katalogi va checkout prototipi.</p></article>
    <article><h3>Tech Blog</h3><p>Article, time, aside va archive jadvali.</p></article>
  </main>
  <footer><p>&copy; 2026 Jasur Portfolio</p></footer>
</body>
</html>

```

contact.html

```

<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Jasur | Kontakt</title>
  <meta name="description" content="Jasur bilan bog'lanish formasi.">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Jasur Portfolio</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="about.html">Men haqimda</a> <a href="projects.html">Loyihalar</a> <a
href="contact.html">Kontakt</a></nav>
  </header>
  <main id="main">

    <section><h2>Bog'lanish</h2><form action="/contact" method="post">
      <label for="name">Ism</label><input id="name" name="name" autocomplete="name" required>
      <label for="email">Email</label><input id="email" name="email" type="email"
autocomplete="email" required>
      <label for="message">Xabar</label><textarea id="message" name="message" rows="6"
required></textarea>
      <button type="submit">Yuborish</button>
    </form></section>
  </main>
  <footer><p>&copy; 2026 Jasur Portfolio</p></footer>
</body>
</html>

```

147 JAVOB — Universitet sayti

Loyiha fayllari

```
project/
  index.html
  faculties.html
  news.html
  admission.html
```

index.html

```
<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Universitet | Bosh sahifa</title>
  <meta name="description" content="Texnologiyalar Universiteti haqida asosiy ma'lumot.">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Texnologiyalar Universiteti</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="faculties.html">Fakultetlar</a> <a href="news.html">Yangiliklar</a> <a
href="admission.html">Qabul</a></nav>
  </header>
  <main id="main">
    <section><h2>Kelajak texnologiyalarini bugun o'rganing</h2><p>Dasturiy injiniring,
kiberxavfsizlik va telekommunikatsiya.</p></section>
    <section><h2>Tezkor havolalar</h2><ul><li><a
href="faculties.html">Fakultetlar</a></li><li><a
href="admission.html">Qabul</a></li></ul></section>
  </main>
  <footer><p>&copy; 2026 Texnologiyalar Universiteti</p></footer>
</body>
</html>
```

faculties.html

```
<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Universitet | Fakultetlar</title>
  <meta name="description" content="Universitet fakultetlari va yo'nalishlari.">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Texnologiyalar Universiteti</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="faculties.html">Fakultetlar</a> <a href="news.html">Yangiliklar</a> <a
href="admission.html">Qabul</a></nav>
  </header>
  <main id="main">
    <h2>Fakultetlar</h2>
    <section><h3>Dasturiy injiniring</h3><p>Web, mobile va software engineering.</p></section>
    <section><h3>Kiberxavfsizlik</h3><p>Tarmoq va axborot xavfsizligi.</p></section>
    <section><h3>Telekommunikatsiya</h3><p>Zamonaviy aloqa tizimlari.</p></section>
  </main>
  <footer><p>&copy; 2026 Texnologiyalar Universiteti</p></footer>
</body>
</html>
```

news.html

```

<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Universitet | Yangiliklar</title>
  <meta name="description" content="Universitetning so'nggi yangiliklari.">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Texnologiyalar Universiteti</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="faculties.html">Fakultetlar</a> <a href="news.html">Yangiliklar</a> <a
href="admission.html">Qabul</a></nav>
  </header>
  <main id="main">

    <h2>Yangiliklar</h2>
    <article><h3>Yangi laboratoriya ochildi</h3><p><time datetime="2026-09-10">10-sentabr
2026</time></p><p>Talabalar uchun yangi amaliyot muhiti ishga tushdi.</p></article>
    <article><h3>Hackathon natijalari</h3><p><time datetime="2026-09-01">1-sentabr
2026</time></p><p>Talabalar jamoalari innovatsion loyihalarni namoyish etdi.</p></article>
  </main>
  <footer><p>&copy; 2026 Texnologiyalar Universiteti</p></footer>
</body>
</html>

```

admission.html

```

<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Universitet | Qabul</title>
  <meta name="description" content="Universitet qabul dasturlari va ariza formasi.">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Texnologiyalar Universiteti</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="faculties.html">Fakultetlar</a> <a href="news.html">Yangiliklar</a> <a
href="admission.html">Qabul</a></nav>
  </header>
  <main id="main">

    <section><h2>Qabul dasturlari</h2><table><caption>Bakalavr
yo'nalishlari</caption><thead><tr><th scope="col">Yo'nalish</th><th
scope="col">Davomiylik</th></tr></thead><tbody><tr><th scope="row">Dasturiy
injiniring</th><td>4 yil</td></tr><tr><th scope="row">Kiberxavfsizlik</th><td>4
yil</td></tr></tbody></table></section>

    <section><h2>Ariza</h2><form action="/admission" method="post"><fieldset><legend>Shaxsiy
ma'lumotlar</legend><label for="name">Ism</label><input id="name" name="name" required><label
for="email">Email</label><input id="email" name="email" type="email"
required></fieldset><fieldset><legend>Yo'nalish</legend><label
for="program">Dastur</label><select id="program" name="program" required><option
value="">Tanlang</option><option>Dasturiy
injiniring</option><option>Kiberxavfsizlik</option></select></fieldset><button
type="submit">Ariza yuborish</button></form></section>
  </main>
  <footer><p>&copy; 2026 Texnologiyalar Universiteti</p></footer>
</body>
</html>

```

148 JAVOB — Online do'kon

Loyiha fayllari

```
project/
  index.html
  products.html
  product.html
  checkout.html
```

index.html

```
<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Tech Shop | Bosh sahifa</title>
  <meta name="description" content="Texnologik mahsulotlar online do'koni.">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Tech Shop</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="products.html">Mahsulotlar</a> <a href="product.html">Mahsulot</a> <a
href="checkout.html">Checkout</a></nav>
  </header>
  <main id="main">

    <section><h2>Texnologiyani oson tanlang</h2><p>O'qish va ish uchun tanlangan
aksessuarlar.</p><a href="products.html">Katalogni ko'rish</a></section>
    <section><h2>Afzalliklar</h2><ul><li>Aniq tavsif</li><li>Yetkazish
ma'lumoti</li><li>Accessible checkout</li></ul></section>
  </main>
  <footer><p>&copy; 2026 Tech Shop</p></footer>
</body>
</html>
```

products.html

```
<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Tech Shop | Mahsulotlar</title>
  <meta name="description" content="Tech Shop mahsulot katalogi.">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Tech Shop</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="products.html">Mahsulotlar</a> <a href="product.html">Mahsulot</a> <a
href="checkout.html">Checkout</a></nav>
  </header>
  <main id="main">

    <h2>Mahsulotlar</h2>
    <section aria-label="Mahsulot katalogi"><article><h3>Mexanik klaviatura</h3><p><strong>650 000 so'm</strong></p><a
href="product.html">Batafsil</a></article><article><h3>Simsiz sichqoncha</h3><p><strong>280 000
so'm</strong></p></article><article><h3>Quloqchin</h3><p><strong>420 000 so'm</strong></p></article></section>
  </main>
  <footer><p>&copy; 2026 Tech Shop</p></footer>
</body>
</html>
```

product.html

```

<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Mexanik klaviatura | Tech Shop</title>
  <meta name="description" content="Mexanik klaviatura tavsifi va yetkazish ma'lumoti.">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Tech Shop</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="products.html">Mahsulotlar</a> <a href="product.html">Mahsulot</a> <a
href="checkout.html">Checkout</a></nav>
  </header>
  <main id="main">

    <article><h2>Mexanik klaviatura</h2><figure><figcaption>87 tugmali mexanik
klaviatura</figcaption></figure><p><strong>650 000
so'm</strong></p><h3>Xususiyatlar</h3><ul><li>USB-C</li><li>Hot-swap
switch</li><li>O'zbek/ingliz layout sticker</li></ul><h3>Yetkazish</h3><table><tr><th
scope="col">Hudud</th><th scope="col">Muddat</th></tr><tr><th scope="row">Toshkent</th><td>1
kun</td></tr><tr><th scope="row">Viloyatlar</th><td>2-4 kun</td></tr></table><button
type="button">Savatga qo'shish</button></article>
  </main>
  <footer><p>&copy; 2026 Tech Shop</p></footer>
</body>
</html>

```

checkout.html

```

<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Checkout | Tech Shop</title>
  <meta name="description" content="Buyurtma ma'lumotlarini kiritish.">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Tech Shop</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="products.html">Mahsulotlar</a> <a href="product.html">Mahsulot</a> <a
href="checkout.html">Checkout</a></nav>
  </header>
  <main id="main">

    <h2>Checkout</h2><form action="/order" method="post"><fieldset><legend>Aloqa</legend><label
for="name">Ism</label><input id="name" name="name" autocomplete="name" required><label
for="email">Email</label><input id="email" name="email" type="email" autocomplete="email"
required><label for="tel">Telefon</label><input id="tel" name="tel" type="tel"
autocomplete="tel" required></fieldset><fieldset><legend>Yetkazish</legend><label
for="city">Shahar</label><input id="city" name="city" required><label
for="address">Manzil</label><textarea id="address" name="address"
required></textarea></fieldset><button type="submit">Buyurtmani tasdiqlash</button></form>
  </main>
  <footer><p>&copy; 2026 Tech Shop</p></footer>
</body>
</html>

```

149 JAVOB — Blog platformasi

Loyiha fayllari

```
project/
  index.html
  post.html
  archive.html
  about.html
```

index.html

```
<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Tech Blog | Bosh sahifa</title>
  <meta name="description" content="HTML va web bo'yicha amaliy maqolalar.">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Tech Blog</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="post.html">Maqola</a> <a href="archive.html">Arxiv</a> <a
href="about.html">Muallif</a></nav>
  </header>
  <main id="main">

    <section><h2>So'nggi maqolalar</h2><article><h3><a href="post.html">HTMLni tez
o'rganish</a></h3><p><time datetime="2026-09-12">12-sentabr 2026</time></p><p>Amaliy mashq va
debugging odatlari.</p></article><article><h3>Semantic HTML</h3><p><time datetime="2026-09-
05">5-sentabr 2026</time></p><p>Ma'noli markupning foydasi.</p></article></section>

    <aside><h2>Newsletter</h2><form action="/subscribe" method="post"><label
for="email">Email</label><input id="email" name="email" type="email" required><button
type="submit">Obuna bo'lish</button></form></aside>
  </main>
  <footer><p>&copy; 2026 Tech Blog</p></footer>
</body>
</html>
```

post.html

```
<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>HTMLni tez o'rganish | Tech Blog</title>
  <meta name="description" content="HTMLni amaliy mashqlar orqali o'rganish tavsiyalari.">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Tech Blog</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="post.html">Maqola</a> <a href="archive.html">Arxiv</a> <a
href="about.html">Muallif</a></nav>
  </header>
  <main id="main">

    <article><header><h2>HTMLni tez o'rganish</h2><p><time datetime="2026-09-12">12-sentabr
2026</time></p></header><figure><figcaption>Kichik, muntazam amaliyot.</figcaption></figure><p>HTMLni o'rganishda
tuzilmani tushunish va har kuni kichik sahifa yaratish samarali.</p><h3>Debugging
odati</h3><p>Validator va DevTools yordamida xatoni sabab bilan toping.</p></article>

    <aside><h2>O'xshash mavzular</h2><ul><li><a href="archive.html">Semantic
HTML</a></li><li><a href="archive.html">Formalar</a></li></ul></aside>
  </main>
  <footer><p>&copy; 2026 Tech Blog</p></footer>
</body>
</html>
```

archive.html

```

<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Arxiv | Tech Blog</title>
  <meta name="description" content="Tech Blog maqolalari arxivi.">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Tech Blog</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="post.html">Maqola</a> <a href="archive.html">Arxiv</a> <a
href="about.html">Muallif</a></nav>
  </header>
  <main id="main">

    <h2>Maqolalar arxivi</h2><table><caption>2026 maqolalari</caption><thead><tr><th
scope="col">Sana</th><th scope="col">Maqola</th><th
scope="col">Mavzu</th></tr></thead><tbody><tr><th scope="row">12.09.2026</th><td><a
href="post.html">HTMLni tez o'rganish</a></td><td>HTML</td></tr><tr><th
scope="row">05.09.2026</th><td>Semantic HTML</td><td>Semantika</td></tr></tbody></table>
  </main>
  <footer><p>&copy; 2026 Tech Blog</p></footer>
</body>
</html>

```

about.html

```

<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Muallif | Tech Blog</title>
  <meta name="description" content="Tech Blog muallifi haqida.">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Tech Blog</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="post.html">Maqola</a> <a href="archive.html">Arxiv</a> <a
href="about.html">Muallif</a></nav>
  </header>
  <main id="main">

    <article><h2>Muallif haqida</h2><figure><figcaption>Web dasturlash talabasi</figcaption></figure><p>Amaliy web
darsliklari va maqolalar yozaman.</p><address>Email: <a
href="mailto:hello@example.com">hello@example.com</a></address></article>
  </main>
  <footer><p>&copy; 2026 Tech Blog</p></footer>
</body>
</html>

```

150 JAVOB — FINAL - Nova Studio

Loyiha fayllari

```

project/
  index.html
  about.html
  services.html
  gallery.html
  contact.html

```

index.html

```

<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Nova Studio | Raqamli xizmatlar</title>
  <meta name="description" content="Web, dizayn va raqamli kontent xizmatlari.">
  <meta name="robots" content="index, follow">
  <meta property="og:type" content="website">
  <link rel="canonical" href="https://example.com/">
  <meta property="og:title" content="Nova Studio">
  <meta property="og:description" content="Raqamli xizmatlar">
  <meta property="og:image" content="https://example.com/images/og-cover.jpg">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Nova Studio</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="about.html">Biz haqimizda</a> <a href="services.html">Xizmatlar</a> <a
href="gallery.html">Galereya</a> <a href="contact.html">Kontakt</a></nav>
  </header>
  <main id="main">

    <section><h2>G'oyangizni raqamli mahsulotga aylantiramiz</h2><p>Semantic web-sayt, dizayn
va raqamli kontent.</p><a href="services.html">Xizmatlarni ko'rish</a></section>
    <section><h2>Asosiy xizmatlar</h2><article><h3>Web-sayt</h3><p>Tez va ma'noli HTML
struktura.</p></article><article><h3>Dizayn</h3><p>Mahsulot uchun aniq vizual
tizim.</p></article></section>
  </main>
  <footer><p>&copy; 2026 Nova Studio</p></footer>
</body>
</html>

```

about.html

```

<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Biz haqimizda | Nova Studio</title>
  <meta name="description" content="Nova Studio jamoasi va tarixi.">
  <meta name="robots" content="index, follow">
  <meta property="og:type" content="website">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Nova Studio</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="about.html">Biz haqimizda</a> <a href="services.html">Xizmatlar</a> <a
href="gallery.html">Galereya</a> <a href="contact.html">Kontakt</a></nav>
  </header>
  <main id="main">

    <article><h2>Biz haqimizda</h2><p>Nova Studio <time datetime="2024">2024-yildan</time>
raqamli mahsulotlar ustida ishlaydi.</p><figure><figcaption>Mahsulot, dizayn va web
jamoasi</figcaption></figure><h3>Qadriyatlar</h3><ul><li>Accessibility</li><li>Tezlik</
li><li>Tushunarli kontent</li></ul></article>
  </main>
  <footer><p>&copy; 2026 Nova Studio</p></footer>
</body>
</html>

```

services.html

```

<!DOCTYPE html>
<html lang="uz">

```

```

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Xizmatlar | Nova Studio</title>
  <meta name="description" content="Nova Studio web va dizayn xizmatlari.">
  <meta name="robots" content="index, follow">
  <meta property="og:type" content="website">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Nova Studio</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="about.html">Biz haqimizda</a> <a href="services.html">Xizmatlar</a> <a
href="gallery.html">Galereya</a> <a href="contact.html">Kontakt</a></nav>
  </header>
  <main id="main">

    <h2>Xizmatlar</h2><article><h3>Landing page</h3><p>Marketing maqsadiga mos semantic
sahifa.</p></article><article><h3>Korporativ sayt</h3><p>Ko'p sahifali tuzilma va kontent
arxitekturasini.</p></article><table><caption>Xizmatlar taqqoslanishi</caption><thead><tr><th
scope="col">Xizmat</th><th scope="col">Sahifa</th><th
scope="col">Muddat</th></tr></thead><tbody><tr><th scope="row">Landing</th><td>1</td><td>5-7
kun</td></tr><tr><th scope="row">Korporativ</th><td>5+</td><td>2-4
hafta</td></tr></tbody></table>
  </main>
  <footer><p>&copy; 2026 Nova Studio</p></footer>
</body>
</html>

```

gallery.html

```

<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Galereya | Nova Studio</title>
  <meta name="description" content="Nova Studio loyihalari galereyasi.">
  <meta name="robots" content="index, follow">
  <meta property="og:type" content="website">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Nova Studio</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="about.html">Biz haqimizda</a> <a href="services.html">Xizmatlar</a> <a
href="gallery.html">Galereya</a> <a href="contact.html">Kontakt</a></nav>
  </header>
  <main id="main">

    <h2>Loyihalar galereyasi</h2><figure><picture><source srcset="images/project-1.webp"
type="image/webp"></picture><figcaption>Mobil ilova landing
page</figcaption></figure><figure><figcaption>Ta'lim
platformasi</figcaption></figure><figure><figcaption>Online
do'kon</figcaption></figure>
  </main>
  <footer><p>&copy; 2026 Nova Studio</p></footer>
</body>
</html>

```

contact.html

```

<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Kontakt | Nova Studio</title>

```

```

<meta name="description" content="Nova Studio bilan bog'lanish va brief yuborish.">
<meta name="robots" content="index, follow">
<meta property="og:type" content="website">
</head>
<body>
  <a href="#main">Asosiy kontentga o'tish</a>
  <header>
    <h1>Nova Studio</h1>
    <nav aria-label="Asosiy navigatsiya"><a href="index.html">Bosh sahifa</a> <a
href="about.html">Biz haqimizda</a> <a href="services.html">Xizmatlar</a> <a
href="gallery.html">Galereya</a> <a href="contact.html">Kontakt</a></nav>
  </header>
  <main id="main">

    <section><h2>Bog'lanish</h2><form action="/contact" method="post"><fieldset><legend>Aloqa
ma'lumotlari</legend><label for="name">Ism</label><input id="name" name="name"
autocomplete="name" required><label for="email">Email</label><input id="email" name="email"
type="email" autocomplete="email" required></fieldset><fieldset><legend>Loyiha</legend><label
for="service">Xizmat</label><select id="service" name="service"><option>Landing
page</option><option>Korporativ sayt</option></select><label for="message">Loyiha
haqida</label><textarea id="message" name="message" rows="6"
required></textarea></fieldset><button type="submit">Yuborish</button></form></section>
    <section><h2>Brief namunasi</h2><object data="files/brief.pdf" type="application/pdf"
width="100%" height="500"><p>PDF ochildi. <a href="files/brief.pdf">Briefni yuklab
oling</a>.</p></object></section>
  </main>
  <footer><p>&copy; 2026 Nova Studio</p></footer>
</body>
</html>

```

TUGATDINGIZ!

150 ta challenge + 5 ta capstone

Endi maqsad - CSS va JavaScriptga o'tishdan oldin bitta real HTML loyihani noldan mustaqil yig'ish, validator va accessibility checklist bilan tekshirish.

KEYINGI 3 QADAM

1. Bitta HTML capstone ni o'zingizning real mavzuingizga moslab qayta yarating.
2. CSS bilan layout, responsive dizayn va tipografiyani qo'shing.
3. JavaScript qo'shishda semantic HTML va accessibilityni saqlang.

FINAL SELF-CHECK ☐ validator ☐ keyboard ☐ alt/label ☐ broken link ☐ heading/semantic